arXiv:1511.04745v4 [stat.ME] 30 Apr 2025

The matryoshka doll prior – principled multiplicity correction in Bayesian model comparison

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Abstract. This paper introduces a general and principled construction of model space priors with a focus on regression problems. The proposed formulation regards each model as a "local" null hypothesis whose alternatives are the set of models that nest it. Assuming constant odds between any "local" null and its alternatives provides a natural isomorphism of model spaces (like a matryoshka doll), constituting an intuitive way to correct for test multiplicity. This isomorphism yields the Poisson distribution as the unique limiting distribution over model dimension under mild assumptions. We compare this model space prior theoretically and in simulations to widely adopted Beta-Binomial constructions. We show that the proposed prior yields a "just-right" multiplicity correction that induces a desirable complexity penalization profile.

MSC2020 subject classifications: Primary 62F15; secondary 62J99.

Keywords: Model space prior, Bayesian variable selection, Regression models, infinite exchangeability.

1 Introduction

The problem of identifying suitable sets of predictors in regression models of the form

$$\mathbf{y} = \mathbf{X}_0\boldsymbol{\beta}_0 + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \text{where} \quad \boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \mathbf{I}/\tau), \tag{1}$$

has been extensively studied in the statistical literature. In (1), the matrix $\mathbf{X}_0$ corresponds to p_0 covariates that are necessarily included in the mean structure of $\mathbf{y}$ (one of which must be the intercept term). The matrix $\mathbf{X}$ contains the additional p covariates whose importance is to be tested, leading to 2^p models. Given the increasing number of research questions in which p is large or growing with n (especially the case when $p >> n$), there has been a rekindled interest in the selection/testing problem. In this paradigm, it is imperative to identify strategies that yield consistent selection while remaining computationally viable.

In this manuscript we formulate an intuitive and principled strategy to build model space priors that 1) properly account for test multiplicity across the entire model space, and that 2) are computationally viable, even in the massive p scenario. Endowing the

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prior with the ability to suitably control for test multiplicity will implicitly rein in model complexity as p increases.

In the Bayesian selection problem, the posterior learning rates associated with Bayes' factors work in conjunction with the model space prior to produce posterior inference. To elaborate on this point, assume that we are comparing a finite "true" model M to another finite model M' using local priors on the model-specific parameters.[1] If M is not nested in M', then the learning rate of M versus M' is exponential in n, which makes learning M in the pair-wise comparison relatively easy. However, if M is nested in M', then the Bayes' Factor learning rate of M against M' is a power of n, which makes it difficult for M to beat M' even with large sample sizes. Therefore, when considering the entire model space, the combinatorial size of the number of models nesting the true model hinders learning when the number of comparisons is not accounted for (and so model complexity is not properly penalized).

The literature discussing priors over the model space is sparse, but important progress has been made. Ley and Steel (2009) and Scott and Berger (2010) have advocated for Beta-Binomial(a, b) priors on indicator variables for covariate inclusion. In both of these works the authors recommend setting $a = b = 1$, which provides a Bayesian version of a multiplicity correction. Unfortunately, as Johnson and Rossell (2012) note, this multiplicity correction lacks the strength to induce posterior concentration on a finite true model when local priors on the regression coefficients are used. This concentration issue arises because infinitely exchangeable Beta-Binomial priors (i.e., with fixed a and b) place zero mass on finite models as p increases. To correct this undesirable behavior, Wilson et al. (2010) set $a = 1$ and $b = \lambda p$, which yields (a priori) a finite expected number of included covariates as p increases. Alternatively, Castillo et al. (2015) obtain promising selection and reconstruction results by requiring a slightly faster than exponential decrease in the prior probability of the number of covariates in the model. This is achieved by taking $b = p^u$ for some $u > 1$. To impose a penalty on model complexity, the Beta-Binomial priors in Wilson et al. (2010) and Castillo et al. (2015) forgo infinite exchangeability in favor of finite exchangeability.

When one assumes some kind of exchangeability between predictors, then it is natural to think about the construction of model space priors in terms of model complexity. Nevertheless, this approach may lead to unintended (and perhaps undesirable) consequences. The constructions discussed above provide a prior on complexity through the inclusion indicators and a mixing distribution. The prior on complexity from the Beta-Binomial(a, b) exhibits mass loss, the probability of the set of models with k predictors decreases to 0 as p increases. The prior on complexity from the Beta-Binomial(a, p^u) for $u > 1$ exhibits degeneracy, the probability of the global null model increases to 1 as p increases. The Beta-Binomial$(a, \lambda p)$ produces a proper and non-degenerate prior on complexity as p increases (in fact, the limit is a negative binomial distribution), sidestepping the issues of the two other priors, but has the unintended consequence of having false positive control degrade as the true model size increases (see Section 5).

[1] For examples using local priors, see (Zellner and Siow, 1980; Liang et al., 2008; Berger and Pericchi, 1996; Casella et al., 2009; Moreno et al., 2010; Girón et al., 2010; Casella and Moreno, 2012; Womack et al., 2014; Hans, 2010; Castillo et al., 2015). For examples of non-local priors, see (Johnson and Rossell, 2010, 2012; Altomare et al., 2013).

We consider that a better way to build a prior is to define it by enforcing the consequences that we want (i.e., multiplicity control) in the prior, and allowing the prior on model complexity to be implied by this construction. We propose treating each model as a local null hypothesis with the set of models that nest it serving as its corresponding local alternatives. The explicit multiplicity correction that our construction enforces is that the probability of a model and that of the set of its local alternatives are comparable regardless of model size k or total number of predictors p. This induces a proper and non-degenerate prior on model complexity (like the Beta-Binomial$(a, \lambda p)$), but with automatic and universal false positive control as a consequence of its construction.

The rest of the manuscript is structured as follows. In Section 2, we detail the model space prior construction that is produced by using a simple proportionality principle, and discuss the intuition behind our default/recommended hyper-parameter choices. In Section 3 we describe the properties of the prior, showing that the limiting distribution on model dimension is Poisson and demonstrating a self-similarity that motivated naming the prior the *matryoshka doll*. In Section 4, we further discuss this self-similarity and show the ubiquity of a simple generalization of our construction. In Section 5 we compare the behavior of the matryoshka doll prior to the Beta-Binomial based prior constructions mentioned above. In Section 6, we empirically investigate through simulations the posterior performance of the matryoshka doll and of the Beta-Binomial constructions, in both finite model spaces and model spaces with an increasing number of covariates. We close the manuscript with a short discussion of ongoing research extending the proposed prior to more complex model spaces.

2 Local null hypotheses and the matryoshka doll prior

The model space $\mathcal{M}_p$ associated with $\mathbf{X}$ in (1) can be viewed in different ways. For instance, it has been introduced as the powerset of $\{1, \ldots, p\}$ and is isomorphic to $\{0, 1\}^{\times p}$, the set of p-dimensional binary vectors. This binary space corresponds to the space generated by all possible binary vectors made up of inclusion indicator variables γ_j assigned to each $j \in \{1, \ldots, p\}$. In previous constructions of model space priors, the γ_j are viewed as random variables and given a prior distribution, inducing a prior on the model space. While this seems reasonable, it requires considerable prior elicitation and specification on the model space to obtain sensible prior behavior, which is typically assessed through the prior odds between models or groups of models.

Here, we opt for the opposite approach, first defining a guiding principle and then building the prior following this principle. Our guiding principle is to control how each model, viewed as a local null hypothesis, relates to the set of models that nest it (i.e., its local alternative). We accomplish this by defining the prior probability of each model as a function of the prior probability of the set of models that nest it. This principle provides a consistent and straightforward multiplicity correction.

2.1 Definition of the matryoshka doll prior

Formally, let M_A for $A \subseteq \{1, \ldots, p\}$ denote the model with mean structure defined by $\mathbf{X}_0\boldsymbol{\beta}_0 + \mathbf{X}_A\boldsymbol{\beta}_A$, where $\mathbf{X}_A$ is the submatrix of $\mathbf{X}$ containing the columns indexed by $a \in A$. Model M_A comes equipped with a prior distribution $\pi(\boldsymbol{\beta}_0, \boldsymbol{\beta}_A, \tau_A | M_A)$, where τ_A is the precision parameter corresponding to M_A. The model space $\mathcal{M}_p$ consists of models M_A for all possible $A \subseteq \{1, \ldots, p\}$. A model M_A is said to be nested in M_B ($M_A \subset M_B$) whenever $A \subsetneq B$. Though (1) corresponds to Gaussian regression, the model space prior constructed in this paper applies to regression models in general.

Given that the prior odds are the metric typically used to evaluate prior behavior, the simple rule we propose for building the prior is to assume that the relative odds for any model $M_A \in \mathcal{M}_p$ (i.e., the local null hypothesis) with respect to the set of models that nest it $\{M_C : C \subseteq \{1, \ldots, p\} \text{ with } A \subsetneq C\}$ (i.e., the corresponding local alternative) is a fixed quantity, which we denote by η. That is, for any $A \subsetneq \{1, \ldots, p\}$

$$\eta = \frac{P(M_A | \mathcal{M}_p, \eta)}{\sum_{\substack{C \subseteq \{1,\ldots,p\} \\ A \subsetneq C}} P(M_C | \mathcal{M}_p, \eta)}, \tag{2}$$

which completely determines the recursion needed to calculate model prior probabilities for all models in $\mathcal{M}_p$, given by

$$P(M_A | \mathcal{M}_p, \eta) = \eta \sum_{\substack{C \subseteq \{1,\ldots,p\} \\ A \subsetneq C}} P(M_C | \mathcal{M}_p, \eta), \tag{3}$$

for all models $M_A \in \mathcal{M}_p$, with $A \subsetneq \{1, \ldots, p\}$. Our construction is completed by applying this rule uniformly over all p. A simple consequence of this is a straightforward calculation of the probability of the global null hypothesis provided below.

Proposition 1. *For all $p \in \mathbb{N}$ we have $P(M_\emptyset | \mathcal{M}_p, \eta) = \eta/(1+\eta)$.*

Proof. The proof follows directly from (3).

$$\begin{aligned} 1 &= \sum_{M \in \mathcal{M}_p}^{p} P(M | \mathcal{M}_p, \eta) \\ &= P(M_\emptyset | \mathcal{M}_p, \eta) + \sum_{M \neq M_\emptyset} P(M | \mathcal{M}_p, \eta) \\ &= P(M_\emptyset | \mathcal{M}_p, \eta) + P(M_\emptyset | \mathcal{M}_p, \eta)/\eta \end{aligned}$$

from which the claim follows. □

Simply put, (3) states that the probability of a model is proportional to the sum of the probabilities of all the models that nest it, with the proportionality constant given by η. In practice, all that is needed to calculate the prior probabilities over the entire model space is selecting a value for η (the prior odds of a model against all models that

nest it) and using the formula recursively. Alternative multiplicity corrections to (3), including one induced by controlling the odds of a model versus its children models (i.e., nesting models with only one additional predictor), are discussed in Section 4.

To understand how the recursion is carried out, it helps to represent the model space as a directed acyclic graph, where the nodes correspond to models, and the edges between them define their nesting relationships (e.g., see the left graph in Figure 1(d)). Then, starting from the largest element in the graph (i.e., the full model), it is trivial to calculate the prior probability for each node by assigning to each of the edges of the graph a probability according to expression (3).

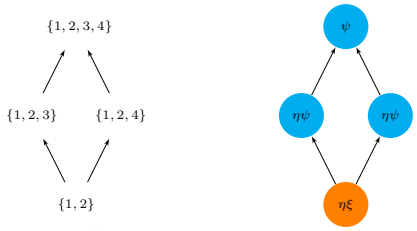
To fix ideas, consider the model space made up by the sixteen models obtained from all possible model combinations available with four predictors, with the intercept only model being the smallest model in the space. In this model space, there are five types of models determined by their complexity (i.e., models with 0, 1, 2, 3 or 4 terms). Given the symmetry of this model space, models of a particular type all have equal prior probability. As such, calculating the probability of one model per model type suffices to determine the probability distribution over the entire model space. In Figure 1 we step through the recursive process (powered by our simple rule) used to calculate the prior over the model space, where the probability in each orange-colored node, corresponds to a proportion of the sum of the probabilities in the blue nodes that sit above them.

Here we show the first steps of the recursion to provide the reader with some intuition for how the prior calculation following (3) proceeds. First, assign a value to η (see Section 2.2 for some guidance), and denote by $\psi = \Pr(M_{\{1,2,3,4\}}|\mathcal{M}_4, \eta)$ the prior probability of the full model. Going down the graph (Figure 1(d)) note that model $M_{\{1,2,3\}}$ is only nested in $M_{\{1,2,3,4\}}$, hence our simple proportionality rule implies that $\Pr(M_{\{1,2,3\}}|\mathcal{M}_4, \eta) = \eta\psi$. Now, because $M_{\{1,2\}}$ is exclusively nested in models $M_{\{1,2,3\}}$, $M_{\{1,2,4\}}$, and $M_{\{1,2,3,4\}}$, its prior is given by $\Pr(M_{\{1,2\}}|\mathcal{M}_4, \eta) = \eta(\eta\psi + \eta\psi + \psi) = \eta\psi(2\eta + 1) = \eta\xi$. The remaining steps of the recursion proceed in a similar fashion and are illustrated in Figure 1. Note that computation of the prior should not use the recursion explicitly. Instead, calculations should be done using the relationships (see (8)) that stem from the prior's isomorphism, which provide superior computational performance.

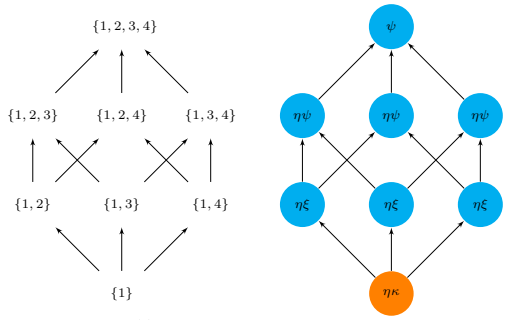
In general, the rule proposed in (3) is quite simple, intuitive and only uses the structure of the model space induced by the nesting of models. The subset relation, $\subsetneq$, induces a partial ordering on $\mathcal{M}_p$ and the prior probability of a model is required to be proportional to the sum of the probabilities of the models that are greater than it in the partial order. Enforcing this proportionality throughout induces a multiplicity correction against local alternatives that leads to a more balanced behavior as we will evidence in later sections of the paper. The rule is also generalizable to any model space with a poset structure. For a given η, the model space prior is uniquely defined whenever the poset has a greatest element (i.e. a full model that nests all other models in $\mathcal{M}_p$). When the poset does not have a unique greatest element, the relative probabilities of the set of greatest elements must be also be set to define the model space prior. An example of the latter would be a regression model space with p possible predictors that places an upper bound (strictly less than p) on the number of predictors allowed in a model. That being said, here we restrict our discussion to properties of the prior on $\mathcal{M}_p$ induced by (3).



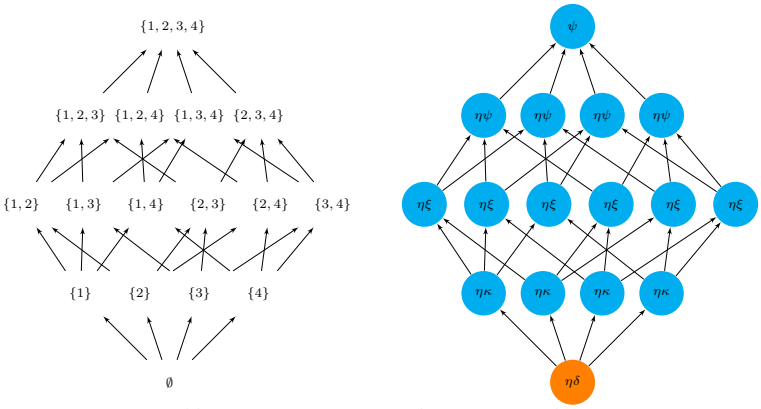
(a) Subgraph for local null $M_{\{1,2,3\}}$



(b) Subgraph for local null $M_{\{1,2\}}$



(c) Subgraph for local null $M_{\{1\}}$



(d) Subgraph for local null $M_{\{\emptyset\}}$ (also represents $\mathcal{M}_p$)

Figure 1: Recursive calculation for the prior probability of one model of each type in the model space. Above $\xi = 2(\eta\psi) + \psi$, $\kappa = 3(\eta\xi) + 3(\eta\psi) + \psi$, and $\delta = 4(\eta\kappa) + 6(\eta\xi) + 4(\eta\psi) + \psi$. The probability for the full model, ψ, is obtained through normalization.

2.2 Choice of η

In order to meaningfully discuss the choice of η, it is useful to preview a few results from later in the paper. First, Theorem 5 shows that as p grows, the induced prior distribution on model complexity has a limiting Poisson distribution with mean $\theta = \log(1 + 1/\eta)$. Second, in Section 3.3 we demonstrate that the convergence (of both the expectation and the distribution) is surprisingly fast and that the prior can be efficiently computed.

These results would apparently provide straightforward means and intuition to specify η. For example, an approach to elicit a fixed η could be through the limiting expectation $\theta = \log(1 + 1/\eta)$. Alternatively, the limiting Poisson distribution on model complexity could guide the specification of a prior on θ that can then be used to induce a prior on $\eta = 1/(\exp(\theta) - 1)$. Furthermore, the speed of prior computation would allow one to learn η through a stochastic algorithm. These things (and more!) can all be done and should not be outside of the abilities of any Bayesian. However, we implore the reader to eschew their Bayesian nature and not think of η as a parameter that is to be learned. Doing so quickly leads one to accidentally negate the prior's intended multiplicity correction.

To illustrate these concerns, suppose that we have $p = 1000$ predictors and *a priori* we think that there should be about 10 meaningful predictors. We could set $\theta = 10$ to match this expectation, which would be equivalent to setting $\eta = 1/(\exp(10) - 1) \approx 4.5 \times 10^{-5}$. Values for η this small are ludicrous in terms of multiplicity correction and could produce undesirable posterior behavior. To further elaborate, assume that 10 of the p predictors are strong, so we would learn them very quickly and be placed in the situation where the model containing just these 10 predictors is a local null hypothesis. In such a case, the prior odds of this model against the models nesting it is 4.5×10^{-5}, strongly promoting the inclusion of false positives. In order to get the same posterior learning rate against a single false positive obtained with $\theta = 1$, we would need to increase the sample size by a factor of 100.

Therefore, to avoid this type of undesirable behavior, we urge the reader to instead simply think of η as it was originally conceived, as the prior odds of any local null hypothesis against all of its alternatives. Similarly, the quantity $\theta = \log(1+1/\eta)$ should not be understood through the limiting Poisson expectation, but rather θ should be thought of as the limiting prior odds of the set of children models to their local null hypothesis (see Table 2 in Section 5.2). Reasonable values for η (or θ) drastically differ depending on whether one thinks of them in terms of prior odds or in terms of prior expected complexity.

This brings us to the specification we recommend for η. Setting $\eta > 1$ (i.e, giving more weight to the local null) does help in posterior inference for learning a local null hypothesis versus its alternatives. One could even go through a laborious frequentist matching exercise (for a given sample size) using the standard posterior odds decision rule. However, we advocate that the reader use either $\eta = 1$ or $\theta = 1$ (corresponding to a $\theta = \log(2)$ and $\eta = 1/(e - 1)$, respectively) as suitable default values. They are both simple, clean, easily interpreted, and allow the structure of matryoshka doll prior to adequately express the multiplicity correction (and, in turn, the implied complexity penalization).

3 Properties of the matryoshka doll prior

In this section, we provide general properties of the prior induced by the rule given in (3) and identify the limiting distribution of the prior.

3.1 Finite exchangeability and conditioning isomorphism

Before going into details, we describe the results briefly.

First, for each p and integer $0 \leq k \leq p$, the set of models M_A with $|A| = k$ forms a complexity class and each model in the class has the same prior probability. For $|A| = k \leq p$ we can write $P(M_A|\mathcal{M}_p, \eta) = \pi_p(k|\eta)\binom{p}{k}^{-1}$, where $\pi_p(k|\eta) = P(|A| = k|\mathcal{M}_p, \eta)$ is the prior on model complexity. This specification is equivalent to finite exchangeability on the inclusion indicators and is quite natural *a priori*.

Second, if we let $B \subsetneq \{1, \ldots, p\}$ and condition on models M_A such that $A \supseteq B$, then the conditional prior $P(M_A|A \supseteq B, \mathcal{M}_p, \eta)$ behaves as if we had moved the covariates corresponding to B into $\mathbf{X}_0$. That is, the conditional prior is isomorphic to the prior that would have been constructed on a model space with $p - |B|$ test covariates. This isomorphism comes directly from the construction in (3) and provides a number of equality constraints on the priors over complexity classes for different values of p.

We now provide the formal statements and proofs. The arguments are essentially about a backwards recursion. If we have p covariates to consider for testing, then we begin with the model space that has no predictors to be tested (all p are assumed to be in the null model). We then add one covariate at a time to the test space. After we have added k covariates to the test space and computed the prior probability on the model space for k covariates, a recursive argument using (3) provides a means of updating model prior probabilities when we add another covariate to the test space. The prior on model complexity is derived through careful book-keeping.

Proposition 2. *Assume that* (3) *holds and let* $\psi_p(\eta) = P\left(M_{\{1,\ldots,p\}} \middle| \mathcal{M}_p, \eta\right)$. *Then the following hold:*

1. *If* $A \subseteq \{1, \ldots, p\}$ *with* $|A| = p - \ell$, *then* $P(M_A|\mathcal{M}_p, \eta) = f_\ell(\eta)\psi_p(\eta)$ *where* $f_0(\eta) = 1$ *and for* $\ell > 0$ *we have* $f_\ell(\eta) = \eta \sum_{j=0}^{\ell-1} f_j(\eta)\binom{\ell}{j}$.
2. *Letting* $\pi_p(k|\eta) = f_{p-k}(\eta)\psi_p(\eta)\binom{p}{k}$ *for* $k = 0, \ldots, p$, *we have* $P(M_B|\mathcal{M}_p, \eta) = \pi_p(k|\eta)\binom{p}{k}^{-1}$ *for* $B \subseteq \{1, \ldots, p\}$ *with* $|B| = k$.
3. *For* $k = 0, \ldots, p-1$, *we have* $\pi_p(k|\eta) = \eta \sum_{j=1}^{p-k} \pi_p(k+j|\eta)\binom{k+j}{k}$.
4. $\psi_p(\eta)^{-1} = \sum_{j=0}^{p} f_{p-j}(\eta)\binom{p}{j}$.

The first claim in Proposition 2 is proven using induction and the rest follows directly from it. The strong result from Proposition 2 is that the probability of a model of dimension $p - \ell$ only depends on p through $\psi_p(\eta)$. This result allows us to make some

nice equality constraints between the probabilities of sets of models of different sizes for differing numbers of potential covariates. The following corollary follows directly from Proposition 2.

Corollary 3. *Let* $\pi_p(k|\eta) = f_{p-k}(\eta)\psi_p(\eta)\binom{p}{k}$ *for* $k = 0, \ldots, p$ *and* $p \in \mathbb{N}$. *Let* $k, \ell,$ *and* j *be integers satisfying* $0 \leq k < p$, $0 < \ell \leq p - k$, *and* $j > 0$, *then*

$$\frac{\pi_p(k+\ell|\eta)}{\pi_p(k|\eta)} = \binom{j+k+\ell}{\ell}\binom{k+\ell}{\ell}^{-1}\frac{\pi_{p+j}(j+k+\ell|\eta)}{\pi_{p+j}(j+k|\eta)}. \tag{4}$$

In particular, we have

$$\frac{\pi_p(j+1|\eta)}{\pi_p(j|\eta)} = \frac{1}{j+1}\frac{\pi_{p-j}(1|\eta)}{\pi_{p-j}(0|\eta)} \tag{5}$$

for $j = 0, 1, 2, \ldots, p-1$.

Proposition 2 also suggests the following isomorphism theorem.

Theorem 4. *Fix* $B \subsetneq \{1, \ldots, p\}$ *with* $|B| = k < p$, *then*

$$P(M_A|A \supseteq B, \mathcal{M}_p, \eta) = \pi_{p-k}(\ell|\eta)\binom{p-k}{\ell}^{-1}. \tag{6}$$

for $B \subseteq A \subseteq \{1, \ldots, p\}$ *with* $|A| = k + \ell \leq p$. *A relative probability preserving isomorphism between the set of models* $\{M_A \in \mathcal{M}_p \ : \ A \supseteq B\}$ *and the model space given by* $\mathcal{M}_{p-k}$ *is obtained by mapping such a set* A *to the set of the ranks of the elements of* $A \setminus B$ *in* $\{1, \ldots, p\} \setminus B$.

Proof. The latter claim follows directly from the former. The former claim can be proven directly using the results of Proposition 2. In particular

$$\begin{aligned} P(M_A|A \supseteq B, \mathcal{M}_p, \eta) &= \frac{P(M_A|\mathcal{M}_p, \eta)}{P(A \supseteq B|\mathcal{M}_p, \eta)} = \frac{f_{p-k-\ell}(\eta)\psi_p(\eta)}{\sum_{j=0}^{p-k} f_{p-k-j}(\eta)\psi_p(\eta)\binom{p-k}{j}} \\ &= \frac{f_{p-k-\ell}(\eta)}{\sum_{j=0}^{p-k} f_{p-k-j}(\eta)\binom{p-k}{j}} = f_{p-k-\ell}(\eta)\psi_{p-k}(\eta) = \pi_{p-k}(\ell|\eta)\binom{p-k}{\ell}^{-1}. \end{aligned}$$

□

The isomorphism theorem motivates naming this prior *the matryoshka doll*. When one reduces the model space by deciding that some variables should be in the base model, one obtains a smaller version of the same model space. Just like opening a matryoshka doll and finding a smaller version of the same doll inside.

A partial converse to Theorem 4 is true. That is, assuming a relative probability preserving isomorphism plus the assumption that $\pi_k(0|\eta) = \eta/(1+\eta)$ for all $k = 1, \ldots, p$ implies expression (3) for a given p. We fully characterize the priors exhibiting the isomorphism condition in Section 4.1.

Notice that Theorem 4 provides an important difference between the matryoshka doll and infinitely exchangeable prior distributions on $\mathcal{M}_p$. Infinite exchangeability arises from an exchangeability and marginalization condition. In essence, we have to envision a universe with an infinite set of potential covariates to include in our models and make our inference on the proposed set of covariates by marginalizing out all but a finite number of the infinite covariates. In contrast, the matryoshka doll asks us to live in a world where the following two reasoners (with the same finite set of p covariates) would reach the same conclusions: *reasoner one* who decides *a priori* to make M_B the base model and *reasoner two* who decides *a posteriori* to condition the analysis on the set of models in $\mathcal{M}_p$ that nest M_B.

3.2 Limiting distribution as p increases

One final implication of expression (3) is that the limit of $\pi_p(k|\eta)$ as p increases is given by $\theta^k \exp(-\theta)/k!$ where $\theta = \log(1+1/\eta)$, which provides a limiting Poisson distribution on model size. The proof relies on the results of Proposition 2 and Corollary 3.

Theorem 5. *Suppose that expression* (3) *holds, then*

$$\lim_{p\to\infty} \pi_p(k|\eta) = \frac{\theta^k \exp(-\theta)}{k!} \tag{7}$$

where $\theta = \log(1 + 1/\eta)$.

Proof. First, from (5), we have

$$\frac{\pi_p(k+1|\eta)}{\pi_p(0|\eta)} = \frac{1}{(k+1)!} \prod_{j=0}^{k} \frac{\pi_{p-j}(1|\eta)}{\pi_{p-j}(0|\eta)}. \tag{8}$$

Letting

$$\theta = \lim_{\ell\to\infty} \frac{\pi_\ell(1|\eta)}{\pi_\ell(0|\eta)}$$

we have

$$\lim_{p\to\infty} \frac{\pi_p(k+1|\eta)}{\pi_p(0|\eta)} = \frac{\theta^{k+1}}{(k+1)!}$$

for each fixed k. We also have

$$1 + 1/\eta = 1 + \sum_{k=0}^{p-1} \frac{\pi_p(k+1|\eta)}{\pi_p(0|\eta)} = 1 + \sum_{k=0}^{p-1} \frac{1}{(k+1)!} \prod_{j=0}^{k} \frac{\pi_{p-j}(1|\eta)}{\pi_{p-j}(0|\eta)}.$$

Applying the Bounded Convergence Theorem and taking the limit inside of the sum establishes

$$1 + 1/\eta = \sum_{j=0}^{\infty} \frac{\theta^j}{j!} = \exp(\theta),$$

which provides $\theta = \log(1 + 1/\eta)$.

□

In Section 5, we compare the limiting Poisson distribution to the limiting distributions obtained by the prior constructions recommended by Scott and Berger (2010), Wilson et al. (2010), and Castillo et al. (2015).

3.3 Prior computation

Computation can be carried out using result 3 from Proposition 2, however it is better to carry out computation using (8) from the proof of Theorem 5. Rewriting (8) as

$$\frac{\pi_p(k+1|\eta)}{\pi_p(1|\eta)} = \frac{1}{(k+1)!}\prod_{j=1}^{k}\frac{\pi_{p-j}(1|\eta)}{\pi_{p-j}(0|\eta)}. \tag{9}$$

provides a quick iterative algorithm for computing $\pi_p(k+1|\eta)/\pi_p(1|\eta)$ for $k = 1, \ldots, p-1$ using the computation of the prior for model spaces with $j = 1, \ldots, p-1$ predictors. The calculation is completed using the value of $\pi_p(0|\eta)$ from Corollary 1 and the fact that

$$\pi_p(1|\eta) = \frac{1-\pi_p(0|\eta)}{1+\sum_{j=1}^{p-1}\frac{\pi_p(k+1|\eta)}{\pi_p(1|\eta)}}. \tag{10}$$

Another simple consequence of (8) and (9) is an easy computation of the prior expected model complexity. If we let K_p be the random variable on $\{0, \ldots, p\}$ representing model complexity from the matryoshka doll prior, then

$$\mu_p(\eta) = E[K_p|\eta] = \frac{\pi_p(1|\eta)}{\pi_{p-1}(0|\eta)}. \tag{11}$$

Though $\pi_p(0|\eta)$ is constant over p for the matryoshka doll, we leave the expected value formula in this form because it only depends on the isomorphism from Theorem 4. Similarly, the computation of the prior using (9) only depends on the isomorphism and the value of $\pi_p(0|\eta)$. These formulae also hold for the generalized matryoshka doll discussed in Section 4.

p	min	lq	mean	median	uq	max
100	0.01	0.02	0.02	0.02	0.02	0.05
500	0.26	0.29	0.29	0.29	0.29	0.97
1000	0.88	0.97	0.99	0.97	0.98	5.30
5000	22.12	24.20	24.58	24.52	24.82	48.20
10000	88.96	93.23	97.48	94.29	95.18	3018.28
50000	1857.47	1961.20	1976.58	1965.03	1968.45	4915.85

Table 1: Benchmarked computation time in milliseconds (1000 replicates) of the matryoshka doll prior in `C++`. Benchmarking done in `R` using the `Rcpp` and `microbenchmark` packages. Computations done on a Mac Mini (2024 M4 with 16GB memory).

Table 1 shows a summary of computation times (over 1000 replicates) in milliseconds of the prior on model complexity for representative values of p. Once p becomes too large (on the order to 10^5), direct computation of the matryoshka doll prior does become

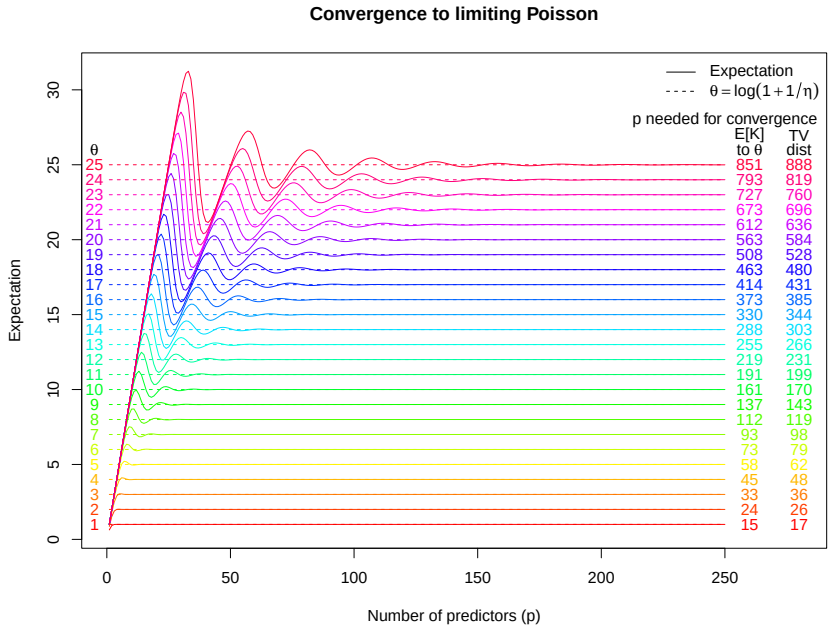


Figure 2: The curves show convergence of the expectation $E[K_p|\eta]$ to the limiting Poisson expectation for $\theta = 1, \ldots, 25$ and $\eta = 1/(\exp(\theta) - 1)$. The first column of numbers on the right is the necessary number of predictors p for the relative difference between the expectation and θ to be less than 1.82×10^{-12}. The second column of numbers on the right is the necessary number of predictors p for the total variation distance between the matryoshka doll and the (truncated) limiting Poisson distribution to be less than 1.82×10^{-12}. Note that we only show large values of θ in order to demonstrate the behavior of prior convergence to the Poisson, but would never advocate setting θ anywhere near 25.

infeasible. However, the fast convergence to the asymptotic Poisson distribution on model complexity prevents this from ever becoming an issue. For example, for one of our default values ($\theta = 1$), convergence to the limiting Poisson in TV norm only requires $p = 17$. Figure 2 shows the convergence of the matryoshka to the limiting Poisson for different values of $\theta = \log(1 + 1/\eta)$.

For reasonable values of θ and p sufficiently large, one can shortcut the computation using (9), by setting $\pi_k(1|\eta)/\pi_k(0|\eta) = \theta$ for sufficiently large k. Alternatively, one could substitute the whole distribution on model complexity from the matryoshka doll with the (truncated) limiting Poisson distribution.

4 Generalized Matryoshka Doll

In order to provide a complete picture of the matryoshka doll prior and its limiting Poisson distribution, in this section we present a generalization of (3) and its equivalence to the isomorphism condition.

4.1 Full characterization of the isomorphism condition

For a finite, non-empty subset D of the natural numbers, let $\mathcal{P}_D$ be the powerset of D. Define $[p] = \{1, \ldots, p\}$ for all natural numbers p. Suppose that $|D| = m$ and define the rank map $\phi_D : \mathcal{P}_D \to \mathcal{P}_{[m]}$ by defining $\phi_D(\{d\}) = \{\text{rank of } d \text{ in } D\}$ for $d \in D$, $\phi_D(\emptyset) = \emptyset$, and $\phi_D(E) = \cup_{e \in E} \phi_D(\{e\})$ for $E \in \mathcal{P}_D$. The Isomorphism Condition states that for any natural number p, a fixed $B \subsetneq [p]$, and every A such that $B \subseteq A \subseteq [p]$ we have

$$P(M_A | A \supseteq B, \mathcal{M}_p, I) = P\left(M_{\phi_{[p]\setminus B}(A)} \middle| \mathcal{M}_{p-|B|}, I\right) \tag{12}$$

where I is any relevant information for prior construction. Note that (12) along with priors on $\mathcal{M}_k$ for $k = 1, \ldots, p-1$ is not enough to uniquely specify a prior distribution on $\mathcal{M}_p$ because there is no way from (12) to determine a probability for $M_\emptyset$ in $\mathcal{M}_p$. In order to complete the construction, we need to specify a sequence of positive real numbers η_k for $k \in \mathbb{N}$ and define

$$P(M_\emptyset | \mathcal{M}_k, I) = \frac{\eta_k}{1 + \eta_k}. \tag{13}$$

This sequence would be subsumed into the relevant information I. These prior probabilities are directly relatable to the success probabilities in the probabilistic forward stepwise representation of distributions on $\mathcal{M}_p$ from Ma (2015).

A weaker requirement than (3) is to allow the proportionality constant to depend on model size in some way. In particular, we define the Proportionality Condition for a sequence of positive real numbers η_k to be

$$P(M | \mathcal{M}_p, I) = \eta_{p-|M|} \sum_{\substack{M' \subseteq [p] \\ M \subsetneq M'}} P(M' | \mathcal{M}_p, I) \tag{14}$$

for $M \subsetneq [p]$. We refer to the priors induced by (14) as generalized matryoshka doll priors and obtain the following theorem.

Theorem 6. *Let η_k be a sequence of positive real numbers. Then (12) and (13) hold if and only if (14) holds. Moreover, in the context of (14), we have $\lim_{k\to\infty} \eta_k = \eta > 0$ if and only if $\lim_{p\to\infty} \pi_p(j|I) = \frac{\theta^j \exp(-\theta)}{j!}$ for fixed j where $\theta = \log(1 + 1/\eta)$.*

The most important implication of Theorem 6 is that if we want some version of the Isomorphism condition to hold, and for the sequence $\pi_p(0|I)$ to converge to a fixed number in the open interval $(0, 1)$, then the limiting probability distribution on model complexity is Poisson. Of course, we can consider other kinds of sequences for $\pi_p(0|I)$,

such as those that converge to 0 or 1 or those that do not have a unique limit as p increases. These give rise to different kinds of limits (or no limit) for the distribution on model complexity. We do not discuss these possibilities in this paper.

4.2 Representations of the generalized matryoshka doll

In considering Theorem 6 and the implications of (12)-(14), it is illuminating to consider the nature of different representations for the same prior distribution on the model space. These representations are driven by the implication of the isomorphism condition in (12) that

$$\pi_p(k|I) = \pi_p(j+k|I)\frac{\pi_{p-k}(0|I)}{\pi_{p-k}(j|I)}\binom{k+j}{j} \tag{15}$$

for $j = 1, \ldots, p-k$. Setting $b_1 + \ldots + b_{p-k} = 1$ provides

$$\pi_p(k|I) = \sum_{j=0}^{p-k} b_j \pi_p(j+k|I)\frac{\pi_{p-k}(0|I)}{\pi_{p-k}(j|I)}\binom{k+j}{j}. \tag{16}$$

As an alternative to (14), one could require the probability of a model, say M in $\mathcal{M}_p$, to be some constant ω_{p-k} times the sum of the probabilities of the models obtained by adding a single covariate to M (the so-called children models of M). Because of the isomorphism condition, this is equivalent to

$$\pi_{p-k}(0|I) = \omega_{p-k}\,\pi_{p-k}(1|I) \tag{17}$$

for $k = 0, \ldots, p-1$.

We get an equivalence between (17) and (14) by choosing

$$b_j = \frac{\frac{1}{j}\,\frac{\pi_{p-k-1}(j-1)}{\pi_{p-k-1}(0)}}{\sum_{j=1}^{p-k}\frac{1}{j}\,\frac{\pi_{p-k-1}(j-1)}{\pi_{p-k-1}(0)}} \quad \text{and} \quad \eta_{p-k} = \frac{\omega_{p-k}}{\sum_{j=1}^{p-k}\frac{1}{j}\,\frac{\pi_{p-k-1}(j-1)}{\pi_{p-k-1}(0)}}. \tag{18}$$

If one chooses $\omega_k = \omega$ in (17) for all k, then a Poisson$\left(\omega^{-1}\right)$ distribution truncated to $\{0, \ldots, p\}$ is immediately obtained for model complexity in $\mathcal{M}_p$ and we get $\eta_p = \left[\sum_{j=1}^{p}\frac{\omega^{-j}}{j!}\right]^{-1}$ for the prior odds of $M_\emptyset$ in $\mathcal{M}_p$.

The representation of the model space prior can be extended using different b_j sequences that sum to one. An illustrative use of this is to make the most generic representation of the isomorphism condition by defining an array $\eta_{i,j}$ for $j = 1, \ldots, i$ and $i = 1, \ldots, p$ and setting

$$\pi_{p-k}(0|I) = \sum_{j=1}^{p-k} \eta_{p-k,j}\,\pi_{p-k}(j|I). \tag{19}$$

This produces the prior induced by (17) with

$$\omega_1 = \frac{\pi_1(0|I)}{1-\pi_1(0|I)} \quad \text{and} \quad \omega_k = \eta_{k,1} + \sum_{j=2}^{k}\left(\frac{\eta_{k,j}}{j!}\prod_{i=k-j+1}^{k-1}\omega_i^{-1}\right) \tag{20}$$

for $k = 2, \ldots, p$ and the prior induced by (14) with

$$\eta_{p-k} = \left[\sum_{j=1}^{p-k} \frac{\prod_{i=1}^{j} \omega_{p-k-j+i}^{-1}}{j!} \right]^{-1}. \tag{21}$$

Any representation of a model space prior satisfying the isomorphism condition provides a generalized matryoshka doll prior with a sequence of probabilities for the null model from (13). The limiting Poisson is inescapable so long as the sequence of η_ks converges to a positive real number.

5 Comparisons to previous constructions

In this section, we compare the properties of the matryoshka doll prior to distributions based on widely used Beta-Binomial constructions (see Scott and Berger, 2010; Wilson et al., 2010; Castillo et al., 2015). The prior probability of the set of models with k covariates is

$$\pi_p^{BB}(k|a,b) = \frac{\Gamma(a+b)\Gamma(p+1)\Gamma(k+a)\Gamma(p-k+b)}{\Gamma(b)\Gamma(a)\Gamma(k+1)\Gamma(p-k+1)\Gamma(p+a+b)}, \tag{22}$$

where the superscript BB denotes the Beta-Binomial prior construction. Similarly, we will use the notation $\pi_p^{MD}(k|\eta)$ to denote the matryoshka doll prior from Section 2. We first compare priors in terms of complexity penalization and then compare them by the associated multiplicity correction across a range of model complexities.

The matryoshka doll prior was defined with explicit multiplicity correction in mind. The complexity penalization that is obtained is a consequence of uniform application of this multiplicity correction. The real problem with the $\text{BB}(a,b)$ for fixed b is mass loss. No fixed complexity class gets positive probability as p increases. The Beta-Binomial constructions with b increasing as a function of p were designed to control model complexity by adding an artificially increasing number of "failures" into the prior. The problem with the $\text{BB}(a,p^u)$ for $u > 1$ is degeneracy. No class with positive complexity gets positive probability as p increases. The comparative advantage of the $\text{BB}(a,\lambda p)$ is that it produces a proper, non-degenerate prior distribution on model complexity. However, as the dimension of the local null hypothesis M increases, the odds of M versus its children (or ancestors) decrease to zero as a function of $|M|$ (see Table 2 and (30)). By assuming constant odds to ancestor (or children) models, the matryoshka doll construction is logically coherent avenue through which Bayesians can think about model space prior specification.

5.1 Limiting behavior of $\pi_p(k)$

Complexity penalization can be investigated through the ratio $\pi_p(k+1)/\pi_p(k)$, which is motivated by Castillo et al. (2015); their detection and reconstruction results rely on

$$c_1 p^{-c_2} \leq \frac{\pi_p(k+1)}{\pi_p(k)} \leq c_3 p^{-c_4}. \tag{23}$$

In general, for a Beta-Binomial prior, we have

$$\frac{\pi_p^{BB}(k+1|a,b)}{\pi_p^{BB}(k|a,b)} = \left(\frac{k+a}{k+1}\right)\left(\frac{p-k}{p-k-1+b}\right). \tag{24}$$

We are interested in the different limits for (24) as p increases for fixed k when b is a function of p. For ease of discussion, we will take $a = 1$.

Proposition 7. *Suppose that k is finite and $a = 1$ in (24), then:*

1. *If b is constant as a function of p, then (24) tends to 1 as p increases.*
2. *If $b = \lambda p$, then (24) tends to $1/(\lambda + 1)$ as p increases.*
3. *If $b = p^u$ for $u > 1$, then (24) is approximated by p^{1-u} as p increases.*

In more detail, Proposition (7) implies that if b is constant then the prior on model complexity exhibits mass loss; the prior probability of models with dimension less than any fixed k decreases like k/p as p increases. Also, taking $b = \lambda p$ provides some penalization and a limiting geometric distribution on model complexity. Unfortunately, this penalization is not enough to get posterior concentration on the true model. The prior does not penalize enough to reduce false positives as the size of the true model increases, which motivated Castillo et al. (2015) to require (23). However, if $b = p^u$ for $u > 1$ then the prior degenerates to a point mass on the empty model. This produces the unfortunate behavior of controlling false positives too strongly and creating false negatives for weak signals.

In contrast, the matryoshka doll prior provides

$$\lim_{p\to\infty} \frac{\pi_p^{MD}(k+1|\eta)}{\pi_p^{MD}(k|\eta)} = \lim_{p\to\infty} \frac{1}{k+1}\frac{\pi_{p-k}(1|\eta)}{\pi_{p-k}(0|\eta)} = \frac{\theta}{k+1}, \tag{25}$$

where $\theta = \log(1 + 1/\eta)$, which is an adaptive version of (23). The limiting additional penalization for adding a covariate does not depend on the total number of covariates under consideration, rather it depends on local model complexity. When k increases to be a power of p, then (23) holds for the matryoshka doll prior, implying a penalty adaptation that meets the requirements specified in Castillo et al. (2015) without degenerating to a single point mass as p increases. In this sense, the matryoshka doll prior is a "just right" prior on model complexity (perhaps even Goldilocks would approve).

5.2 Comparison to children models

In the variable selection problem the hardest models for the true model to beat are those with all of the true positives and just one false positive. These are the children models of the true model in the poset of models. Thus, studying the prior odds between a model M and a model $M' \supset M$ with $|M| + 1$ predictors provides insights about the effect of the model space prior on posterior inference.

Proposition 8. *Let M be a model with complexity k and define the children set of models to be $\mathcal{C}(M) = \{M' \supset M \text{ and } |M'| = |M| + 1\}$. For $M' \in \mathcal{C}(M)$, the prior odds under any finitely exchangeable prior is*

$$odds(M', M) = \frac{P(M'|\mathcal{M}_p)}{P(M|\mathcal{M}_p)} = \frac{\pi_p(k+1)}{\binom{p}{k+1}} \frac{\binom{p}{k}}{\pi_p(k)} = \frac{k+1}{p-k} \times \frac{\pi_p(k+1)}{\pi_p(k)} \tag{26}$$

and the prior odds versus the set of children models is

$$odds(\mathcal{C}(M), M) \;=\; \sum_{M' \in \mathcal{C}(M)} odds(M', M) \;=\; (k+1) \times \frac{\pi_p(k+1)}{\pi_p(k)}. \tag{27}$$

Under the matryoshka doll, $odds(\mathcal{C}(M), M)$ is given by

$$odds^{MD}(\mathcal{C}(M), M \,|\, \eta) = \mu_{p-k}(\eta) \times \frac{\pi_{p-k-1}(0|\eta)}{\pi_{p-k}(0|\eta)} \tag{28}$$

where $\mu_{p-k}(\eta)$ is the prior expected model complexity from (11). Note: we have left these odds in this general form that extends to the generalized matryoshka doll even though $\pi_p(0|\eta)$ is constant for the matryoshka doll.

Table 2 displays the values for (26) and (27) attained by the different prior constructions and limiting situations for model complexity. We investigate finite model size, model size growing as $\log(p)$, model size growing as p^q for $0 < q < 1$, and model size growing as a proportion of p. Though this list is not exhaustive, it provides enough variation to observe the influence of the prior on posterior inference. There are a couple of important things to note in this table.

First we discuss finite model size, $|M| = k < \infty$. The beta-binomial with $b = p^u$ with $u > 1$ provides $\text{odds}(\mathcal{C}(M), M)$ that converge to 0 at a k/p^{u-1} rate as p increases for any fixed k. The beta-binomial with constant b or $b = \lambda p$ provides $\text{odds}(\mathcal{C}(M), M)$ that converge to a linear function of k as p increases. The slope for constant b is one and the slope for $b = \lambda p$ is less than one. The matryoshka doll provides $\text{odds}(\mathcal{C}(M), M) = \mu_{p-k}$, which converges to θ as p increases.

Second we discuss increasing model size. The beta-binomial odds follow the same patterns as before for each regime for b. Essentially, the k for finite models is replaced with the appropriate growth rate for k as a function of p. As $|M|$ increases with p, this can cause issues with false positives for the beta-binomial with constant b or $b = \lambda p$, and issues with false negatives for $b = p^u$ for $u > 1$. Once again, the matryoshka doll provides $\text{odds}(\mathcal{C}(M), M) = \mu_{p-k}$, which converges to θ as p increases so long as $p-k$ also increases to infinity. Thus, the matryoshka doll provides nearly constant odds versus the set of children models regardless of the growth rate of k with p (up until the case when $p - k$ is bounded above as p increases). Interestingly, the beta-binomial with $b = p^u$ requires $u \geq 2$ to produce the same (or more) penalization as the matryoshka doll when $k \propto p$.

Figure 3 provides heatmaps of the odds of a single child model (top row in each panel) and the set of children models (bottom row) versus a model as a function of

Prior	exact odds	k finite	$\lim \frac{k}{\log(p)} = \xi$	$\lim \frac{k}{p^q} = \xi$	$\lim \frac{k}{p} = \xi$
Limiting prior odds of a single child model versus a model					
$MD(\theta)$	$\frac{\mu_{p-k}}{p-k}$	$\frac{\theta}{p}$	$\frac{\theta}{p}$	$\frac{\theta}{p}$	$\frac{\theta}{p(1-\xi)}$
$BB(a,b)$	$\frac{k+a}{p-k-1+b}$	$\frac{k+a}{p}$	$\frac{\log(p)}{p}$	$\frac{\xi}{p^{1-q}}$	ξ
$BB(a,\lambda p)$	$\frac{k+a}{p-k-1+\lambda p}$	$\frac{k+a}{p(1+\lambda)}$	$\frac{\xi \log(p)}{p(1+\lambda)}$	$\frac{\xi}{p^{1-q}(1+\lambda)}$	$\frac{\xi}{1+\lambda}$
$BB(a,p^u)$	$\frac{k+a}{p-k-1+p^u}$	$\frac{k+a}{p^u}$	$\frac{\xi \log(p)}{p^u}$	$\frac{\xi}{p^{u-q}}$	$\frac{\xi}{p^{u-1}}$
Limiting prior odds of the set of children models versus a model					
$MD(\theta)$	μ_{p-k}	θ	θ	θ	θ
$BB(a,b)$	$\frac{(k+a)(p-k)}{p-k-1+b}$	$k+a$	$\xi \log(p)$	ξp^q	$\xi(1-\xi)p$
$BB(a,\lambda p)$	$\frac{(k+a)(p-k)}{p-k-1+\lambda p}$	$\frac{k+a}{1+\lambda}$	$\frac{\xi}{1+\lambda}\log(p)$	$\frac{\xi}{1+\lambda}p^q$	$\frac{\xi(1-\xi)}{1+\lambda}p$
$BB(a,p^u)$	$\frac{(k+a)(p-k)}{p-k-1+p^u}$	$\frac{k+a}{p^{u-1}}$	$\frac{\xi \log(p)}{p^{u-1}}$	$\frac{\xi}{p^{u-1-q}}$	$\frac{\xi(1-\xi)}{p^{u-2}}$

Table 2: Limiting prior odds of children models versus a model with k predictors under different limiting regimes for k as p increases. In this table, $\xi > 0$ is a constant (with $\xi < 1$ if $k \propto p$), $u > 1$, $0 < q < 1$, and $\theta = \log\left(1 + \frac{1}{\eta}\right)$.

model complexity and growth rate of k with respect to p. For these plots, we choose $a = 1$, $u = 2$, and $\lambda = 1$ and set $\theta = 1$ so that the matryoshka doll has the same limiting expectation as the BB$(1, \lambda p)$. Some additional insights can be gleaned from the figure that are harder to understand just using Table 2. First, the beta-binomial with either $b = 1$ or $b = p$ displays an inability to properly penalize the set of children models even when k is finite (panels (a), (b) and (c), bottom row). Second, the beta-binomial with fixed a and b even fails to penalize a single child model when $k = \xi p$ and ξ is away from 0 (panel (d), top row). Third, the beta-binomial with $b = p^2$ over-penalizes the set of children models in all scenarios except when $k = \xi p$ with ξ values away from 0 or 1. As expected from its construction, the matryoshka doll appropriately penalizes the set of children models regardless of the size of the model or its growth rate with p.

5.3 Comparison to nesting models

The explicit multiplicity correction of the matryoshka doll construction provides constant prior odds for a model versus the set of nesting models. In this subsection, we compare this to the prior odds obtained through the various Beta-Binomial prior constructions. Once again, assume a model M of dimension k. The prior odds of the models that nest M versus M is given by

$$\frac{\sum_{M' \supsetneq M} P(M'|\mathcal{M}_p)}{P(M|\mathcal{M}_p)} = \sum_{j=1}^{p-k} \frac{\pi_p(k+j)}{\pi_p(k)} \binom{k+j}{j}. \tag{29}$$

Of course, because of (3), the matryoshka doll prior provides a value of $1/\eta$ for (29) regardless of p and k. When $\pi_p(k)$ comes from the Beta-Binomial(a, b) construction,

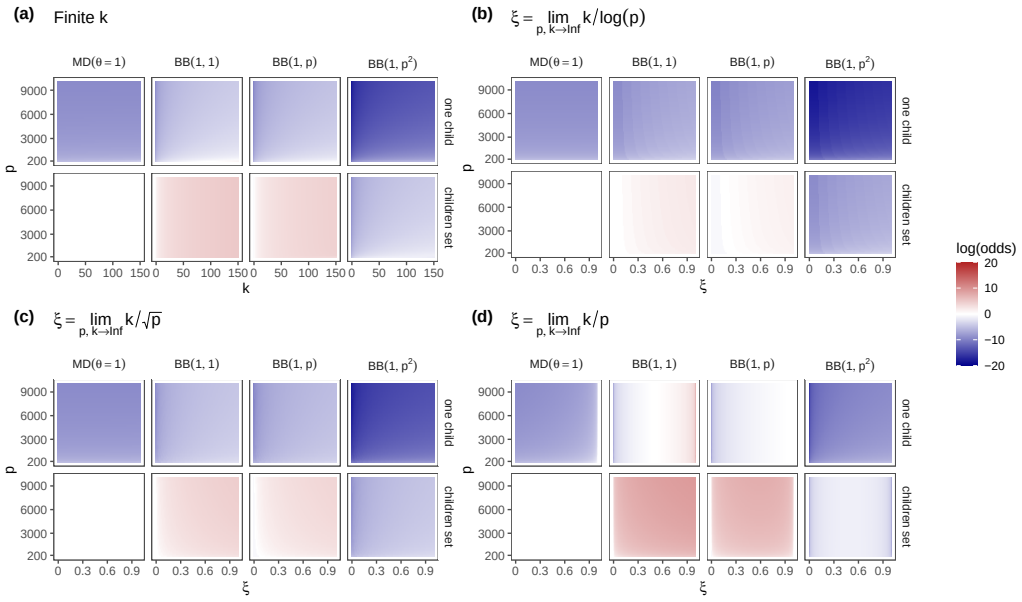


Figure 3: Comparison across model space priors in terms of the odds of a model against a child model (i.e., odds(M', M)) and against the set of children models (i.e., odds$(\mathcal{C}(M), M)$) as k grows with p under different growth regimes, namely: (a) for finite values of k, (b) as k grows proportional to $\log p$, (c) as k grows proportional to $\sqrt{p}$, and (d) as k grows proportional to p.

then (29) is given by

$$\frac{\sum_{M' \supsetneq M} P^{BB}(M'|\mathcal{M}_p, a, b)}{P^{BB}(M|\mathcal{M}_p, a, b)} = \sum_{j=1}^{p-k} \frac{(p-k)!\ \Gamma(k+j+a)\ \Gamma(p-k-j+b)}{j!\ (p-k-j)!\ \Gamma(k+a)\ \Gamma(p-k+b)}. \tag{30}$$

When a and b are constants, the sum in (30) tends to ∞ as p increases for any fixed k, exhibiting the mass loss of the prior. When $b = p^u$ for $u > 1$, (30) tends to 0 as p increases for any fixed k, showing the overly strong penalization of this prior. Further, if $u \geq 2$, then the limit of (30) is 0 even if k increases with p (except in the case when $k \propto p$ and $u = 2$). This overly strong penalization (combined with Bayes' Factor learning rates) leads to undesired false negatives, especially of weak but meaningful predictors.

An interesting limit occurs when $b = \lambda p$. The limit of (30) for finite k is

$$\lim_{p\to\infty} \frac{\sum_{M' \supsetneq M} P^{BB}(M'|\mathcal{M}_p, a, b = \lambda p)}{P^{BB}(M|\mathcal{M}_p, a, b = \lambda p)} = \sum_{j=1}^{\infty} \frac{\Gamma(k+j+a)}{j!\ \Gamma(k+a)} \left(\frac{1}{1+\lambda}\right)^j. \tag{31}$$

The series from (31) can be written in terms of probabilities from a negative-binomial distribution. In particular, if Q_k is a random variable following a negative-binomial distribution that counts the number of successes until $k + a$ failures with individual trial probability of success equal to $1/(1 + \lambda)$, then (31) becomes

$$\lim_{p\to\infty} \frac{\sum_{M' \supsetneq M} P^{BB}(M'|\mathcal{M}_p, a, b = \lambda p)}{P^{BB}(M|\mathcal{M}_p, a, b = \lambda p)} = \frac{P(Q_k > 0)}{P(Q_k = 0)} = \left(1 + \frac{1}{\lambda}\right)^{k+a} - 1. \tag{32}$$

Thus, the prior proposed by Wilson et al. (2010) does have a finite limit for each fixed k. However, that limit increases exponentially as a function of k. This explains the poor posterior concentration as the dimension of the true model increases that was discussed in Castillo et al. (2015).

The matryoshka doll strikes a balance between under- and over- penalizing the set of nesting models. Because (29) is a constant for all k and p, the prior helps the Bayes' factors overcome the combinatorial complexity of the model space without over-penalizing. In Section 6 and in the online supplement, we provide empirical evidence that constant b or $b \propto p$ leads to an inflated number of false positives and lack of concentration of the posterior. Similarly, we provide empirical evidence that the degeneration of the $BB(a, p^u)$ prior to a point mass at the base model controls false positives too aggressively, leading to excessive false negatives and unwarranted posterior concentration on models that exclude covariates that explain meaningful variation in the response variable.

The real problem with the $\text{BB}(a, b)$ for growing p is mass loss. No fixed complexity class gets positive probability as p increases. The real problem with the $\text{BB}(a, p^u)$ for $u > 1$ is degeneracy. No class with positive complexity gets positive probability as p increases. The strength of the $\text{BB}(a, \lambda p)$ is that it produces a proper, non-degenerate prior distribution on model complexity. However, as the dimension of the local null hypothesis M increases, the odds of M versus its children (or ancestors) decrease to zero as a function of $|M|$ (see Table 2 and (30)). The matryoshka doll focuses on the odds as the primary quantity of prior interest. By assuming constant odds to ancestor (or children) models, the matryoshka doll construction is logically coherent avenue through which Bayesians can think about model space prior specification.

6 Performance using synthetic data sets

In order to demonstrate the effects of the multiplicity correction (and ensuing complexity penalization) discussed in previous sections, we now empirically assess the posterior performance of the matryoshka doll prior and compare it to that of the Beta-Binomial constructions. In the simulations we consider the Beta-Binomial(a, b) priors with $a = 1$ and $b \in \{1, p, p^2\}$ and the matryoshka doll prior with $\theta = 1$ ($\eta = 1/(e - 1)$) to match the asymptotic prior expectation of the $\text{BB}(1, p)$ prior. We generate synthetic datasets under multiple scenarios assuming a finite true model. In these experiments we vary the sample size, the number of predictors in the full model, and the rate at which the signal in the predictors decays to better understand how the different priors process signals of varying strength.

6.1 Simulation details

In all of our simulation scenarios the true model M_T is assumed to have $p_T = 5$ predictors. We let $n \in \{41, 81, 121, 161, 201\}$ for scenarios where the sample size varies, while setting the number of predictors in the full model to $p = 20$. In scenarios where we keep the sample size fixed, we set $n = 101$ and let the number of predictors take values in $p \in \{20, 40, \ldots, 100\}$. To control how the signal is assigned to predictors in the true model M_T, we assume the jth largest regression coefficient in the true model, β_j, satisfies $\beta_j^2 \propto \zeta^j$ with $\zeta \in \{1/2, 3/4\}$, such that when $\zeta = 1/2$ the signal is highly concentrated on the first few predictors, whereas with $\zeta = 3/4$, while still decreasing, the signal is more spread out across the true model's coefficients. We choose the value for σ^2 such that the series of coefficients produces the mean structure with signal to noise ratio $\sum \beta_j^2/\sigma^2 = 2$. All Bayes Factors are computed relative to the global null hypothesis (the intercept-only model) using the Zellner-Siow prior on the model parameters with scaling for model complexity (see scaling in equations 8 and 9 of Javier Girón et al. (2006)).

With each combination of n, p, and ζ we simulated 1000 datasets to evaluate the performance of the different model-space prior specifications in terms of: (i) their variable selection ability and (ii) how posterior probability mass concentrates under each prior distribution. For the simulations with fixed $p = 20$, we enumerated the model space. For simulations with varying p, we assessed the posterior model probabilities using a *Pruned Recursion Tree* search algorithm (details in Appendix B).

Briefly: The algorithm begins at the global null model with all predictors available for inclusion. When at a model M, the algorithm creates branches using a subset of $\mathcal{C}(M)$ defined by available predictors. The branches are ordered by decreasing posterior probability. Each branch is a recursion of the algorithm, but with a reduced set of available predictors. Those predictors that were added to M in the higher ranked children models are excluded from the branches defined by the lower ranked children models. The pruning of a branch is done using a deterministic rule. If the relative posterior probability of a child model in its path to the global null model is below a specified cutoff, then the remainder of the corresponding branch is pruned from the search.

For the purposes of this paper, we are using this algorithm to over-search the model space in order to produce posterior inference that we can trust. To achieve this, we use the algorithm with excessively permissive settings for the recursion. We performed the search using a threshold of 0.0001 relative probability for branch pruning. To drive the search, we used the BB$(1, 1)$ model space prior, which further encourages predictor inclusion. The posterior probabilities for the other model space priors shown in our comparisons were computed using renormalization among the set of models found in this permissive search.

Figure 4 shows the number of models found using the algorithm for different values of p and ζ. These numbers, by themselves, are not that useful for determining whether the search has found enough models in the posterior to trust numerical results. To address this, Figure 5 shows the proportion among the models found that is necessary to capture 95% of the re-normalized posterior probability. The only prior that takes a

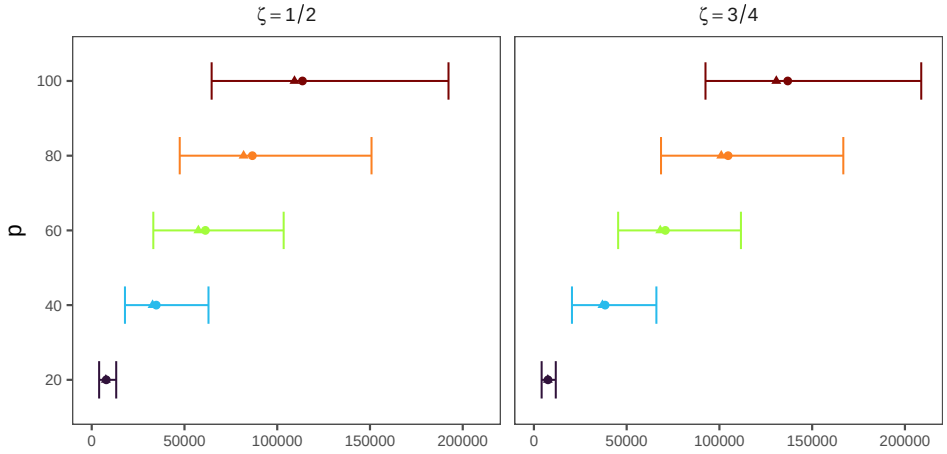


Figure 4: 95% confidence intervals for the number of models found by the search algorithm for different numbers of predictors p and signal decay rate ζ. The mean (●) and the median (▲) are also marked.

substantial portion of the found models to achieve this is the Beta-Binomial$(1, 1)$ with a maximum value of 77.8%. The Beta-Binomial$(1, p)$ and Beta-Binomial$(1, p^2)$ have maximums of 20.3% and 0.3%, respectively. The matryoshka doll with $\theta = 1$ has a maximum of 2.1%. For the latter three priors, the model space search has clearly found an adequate portion of the posterior model space probability mass to be confident with re-normalized posterior inference.

The percentages in Figure 5 foreshadow the results that we will see in terms of posterior concentration as well as true positive and false discovery rates. The Beta-Binomial$(1, p)$ does not concentrate posterior probability enough around the true model and the Beta-Binomial$(1, p^2)$ overly penalizes variable inclusion. The matryoshka doll sits between these priors and provides reasonable posterior concentration while not overly penalizing and decreasing the true positive rate.

We also include results for the Beta-Binomial$(1, 1)$ prior in the empirical studies. However, one could (not unreasonably) argue that an insufficient portion of the posterior model space probability has been explored to feel comfortable with re-normalized posterior inference. While we could run the algorithm with an even more permissive branch-wise stopping rule, we chose not to do so because it has already been well-established that the Beta-Binomial$(1, 1)$ prior is inappropriate when p is large (Johnson and Rossell, 2012). We include the results for completeness, but do not discuss them in the same detail that we discuss the other three priors in.

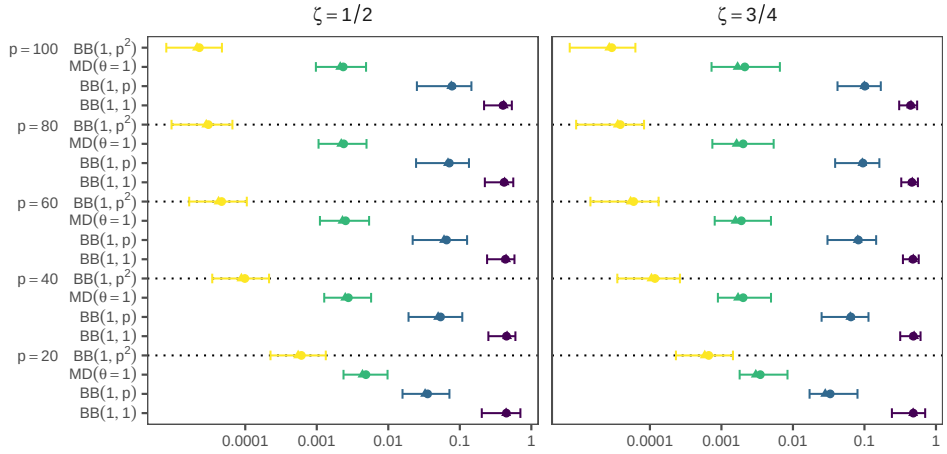


Figure 5: 95% confidence intervals for the percentage of models needed to comprise 0.95 probability out of all models found by the search algorithm for different prior types as well as numbers of predictors p and signal decay rate ζ. The mean (•) and the median (▲) are also marked. The x-axis is presented in the $\log_{10}$ scale to demonstrate the order-of-magnitude difference in the percentage of models needed over the different priors.

6.2 Predictor selection

The main goal of any multiplicity correction is to provide a control on the family-wise false positive rate of a set of decision problems. In order to explore this in the Bayesian framework, we calculated the posterior expected true positive rates (tpr) and false discovery rates (fdr) under each prior for each simulation. These metrics aim to describe how well each of the priors lead the posterior distribution to concentrate on sets of *good* models (i.e., those including many true predictors and few false predictors), which is essential to produce suitable model averaging results. We also obtained the tpr and fdr of the highest probability model (HPM or mode hereforth), which inform about the correspondence between the modal and the true models whenever selection is of interest.

Let $M \setminus M_T$ denote the set of terms in M but not in the true model M_T, and $M \cap M_T$ denotes the set of terms in both M and M_T. To approximate the distributions for the mean tpr and the mean fdr, for each of the $\ell = 1, 2, \ldots, 1000$ data sets generated under a particular simulation scenario we calculate

$$\begin{aligned} \overline{\mathsf{tpr}}_\ell &= \sum_{M \in \mathcal{M}} \mathsf{tpr}(M) \cdot p(M|\mathbf{y}_\ell) \text{ with } \mathsf{tpr}(M) = \frac{|M \cap M_T|}{|M_T|}, \text{ and} \\ \overline{\mathsf{fdr}}_\ell &= \sum_{M \in \mathcal{M}} \mathsf{fdr}(M) \cdot p(M|\mathbf{y}_\ell) \text{ with } \mathsf{fdr}(M) = \frac{|M \setminus M_T|}{|M|}, \end{aligned}$$

where $\mathbf{y}_\ell$ represents the response vector from the ℓth data set. Using the 1000 values obtained for these two metrics under a particular scenario, we approximate their corresponding distribution. The distributions for the tpr and fdr for the mode are obtained by identifying the modal model $M_\ell^\star = \mathsf{argmax}_{M\in\mathcal{M}} p(M|\mathbf{y}_\ell)$ and calculating $\mathsf{tpr}(M_\ell^\star)$ and $\mathsf{fdr}(M_\ell^\star)$ for $\ell = 1, 2, \ldots, 1000$.

Growing n and fixed p

The results produced under each simulation scenario for these four metrics are well-aligned with our discussion in Section 5 regarding how the different priors handle multiplicity correction and complexity penalization. Figure 6 displays the results as n ranges from 41 to 201 with $p = 20$.

In general, the prevailing pattern for the Beta-Binomial(a, b) models is as expected. The BB$(1, 1)$ produces the most true positives but has the worst false discovery rate. The BB$(1, p)$ produces less false discoveries at the cost of some true positives. The BB$(1, p^2)$ is excessively conservative, strongly controlling false discoveries and greatly sacrificing the true positive rate. The matryoshka doll strikes a compromise between the BB$(1, p)$ prior and the BB$(1, p^2)$ prior. It produces adequate false discovery control while not overly-penalizing models that might contain weak true signals.

The trends for tpr described above are exacerbated when $n = 41$ under both signal decay regimes. With the BB$(1, p^2)$ prior the distributions for both the mean tpr and the mode's tpr, concentrate closer to zero than the other priors, indicating that this prior places excessive confidence on smaller models which exclude many of the terms in the true model. Conversely, the other three priors suitably propagate the uncertainty that originates from the limited amount of data to the posterior distribution. When $n = 201$ and $\zeta = 1/2$, the dominating mode for tpr under the BB$(1, p^2)$ is around 0.8, whereas the predominant mode under the other three priors is closer to 1. With the same sample size but with slower signal decay of $\zeta = 3/4$, the weakest predictors are substantially stronger than in the $\zeta = 1/2$ scenario and thus make it easier to detect all of the true predictors regardless of the prior.

The distributions for the mean fdr also follow the expected behavior due to the differences in how the priors handle multiplicity correction. Under the BB$(1, p^2)$ prior the mean fdr distribution is tightly concentrated around zero regardless of the sample size, while for the other three priors the distribution gradually concentrates more and more about zero as the sample size increases. As discussed in Section 5, due to it's decreasing odds of a model versus its children as a function of model size, there is an inflated number of false positives with the BB$(1, p)$ prior compared to the matyoshka doll prior. In terms of the mean fdr, the MD$(\theta = 1)$ prior again provides a sensible compromise between the BB$(1, p^2)$ prior, which penalizes complexity excessively, and the BB$(1, p)$, which does not penalize complexity enough. Lastly, the distribution for the mode's fdr is concentrated around zero no matter the sample size, signal decay scenario, or the prior considered (except for the Beta$(1, 1)$). However, under the BB$(1, p)$ we can see evidence of a longer right tail in the fdr distribution when $\zeta = 1/2$ than is seen under the MD$(\theta = 1)$ prior.

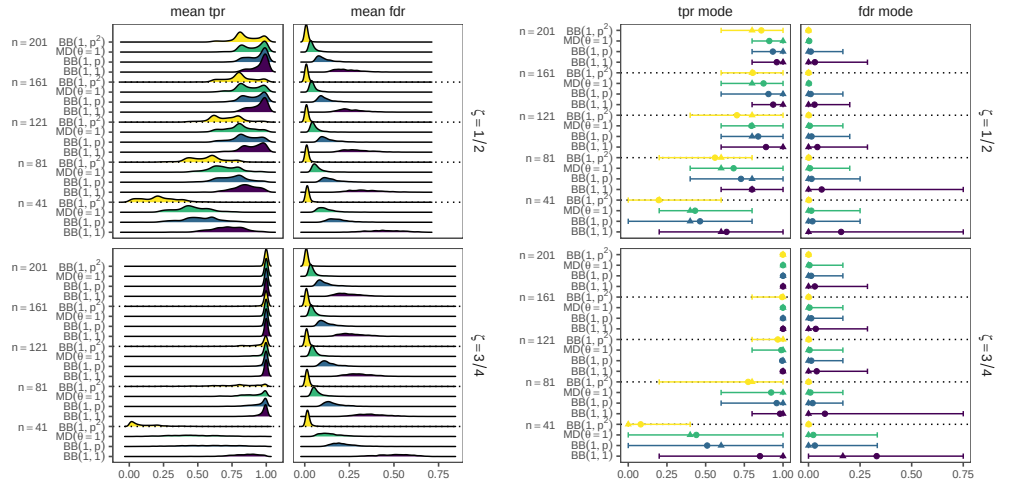


Figure 6: Signal reconstruction metrics for simulation experiments over finite model spaces with growing n and fixed $p = 20$. The figure on the right column displays the 95% confidence intervals, the mean (●) and the median (▲) for tpr mode and fdr mode.

Fixed n and growing p

Figure 7 summarizes the results with $n = 101$ and p taking values in $\{20, 40, 60, 80, 100\}$. In broad terms, the behavior of both the mean tpr and that for the mode are as expected.

As p grows the distribution gradually shifts towards lower values, with the $\text{BB}(1, p^2)$ prior concentrating around lower tpr values followed by the $\text{MD}(\theta = 1)$ and the $\text{BB}(1, p)$ taking the higher values. The distributions under the $\text{MD}(\theta = 1)$ and the $\text{BB}(1, p)$ are very similar, with the modes for the $\text{BB}(1, p)$ happening at higher values than those for the $\text{MD}(\theta = 1)$. It is worth highlighting that the distributions for the tpr metrics when $\zeta = 3/4$ under the $\text{BB}(1, p^2)$ prior become extremely diluted as p grows, providing evidence for how this prior's unduly harsh penalization leads to the exclusion of multiple true signals. Conversely, the other priors are maintain concentrated around tpr values close to one.

In terms of the false discoveries, the mean fdr is tightly concentrated around zero for the $\text{BB}(1, p^2)$. The distribution from the $\text{MD}(\theta = 1)$ prior has most of its mass close to 0.05, with the $\text{BB}(1, p)$ and $\text{B}(1, 1)$ performing the worse, taking mostly fdr values between 0.1 and 0.2 and between 0.2 and 0.45, respectively. Regarding the mode's fdr distributions, these all concentrate around 0 for all priors, but more tightly when $\zeta = 3/4$, indicating that the modal models mostly includes true terms irrespective of the model space prior used. The glaring exception is the $\text{BB}(1, 1)$ whose mode fdr has a long right tail.

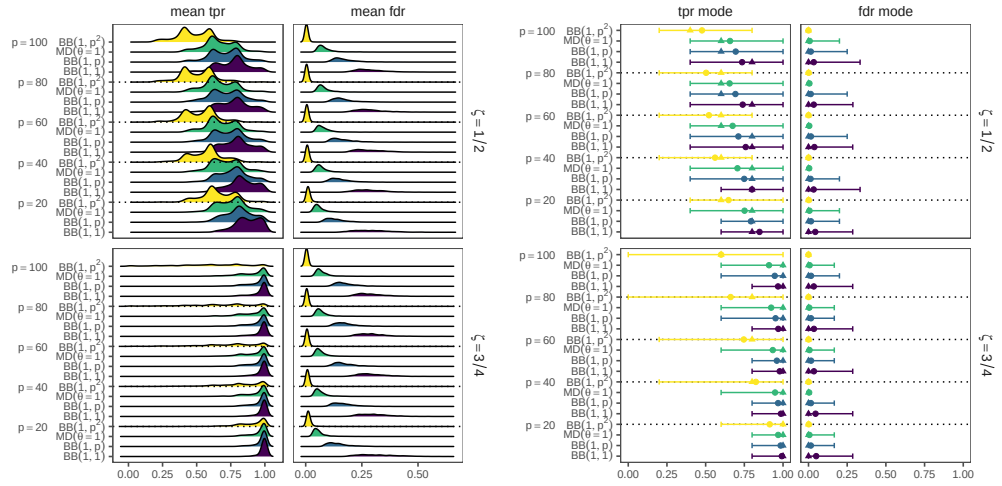


Figure 7: Selection metrics for simulation experiments over finite model spaces with growing p and fixed $n = 101$. The figure on the right displays the 95% confidence intervals, the mean (•) and the median (▲) for tpr mode and fdr mode.

6.3 Posterior Concentration

The tpr and fdr demonstrate the issues of mass loss and degeneracy in the BB$(1, 1)$ and BB$(1, p^2)$ priors, respectively. However, they do not strongly distinguish the BB$(1, p)$ prior from the MD$(\theta = 1)$ prior. The differences in posterior behavior due to these two priors are more easily elucidated using posterior concentration.

We compute four metrics to assess posterior concentration: the posterior probability of the modal model (i.e., the HPM), the posterior probability of the true model, the number of models needed to accumulate 95% of the total probability mass of the models in the space, and the rank for the true model. These are shown in Figure 8 displaying the reuslts for fixed p and growing n and Figure 9 showing the results for growing p and fixed n. The left panel in each figure shows the densities of probability assigned to the modal model and the true model. The right panel in each figure shows that number of models needed to achieve 95% of the probability of the models visited by the search algorithm (with $\log_{10}$ scaled x-axis) and the rank of the true model.

The trends in these figures are quite striking. The BB$(1, p^2)$ produces a posterior distribution that does not properly account for model uncertainty and is generally too confident in a small set of models. It places high probability on the modal model even when it is incorrect and the maximum number of models it needed to achieve 95% of the probability of the visited models was 21 across all scenarios. At the other end of the spectrum is the BB$(1, 1)$ model, which spreads out posterior probabilities across a quite large number of models and places very low probability on the modal model even when p is small and the signal in each predictor is large.

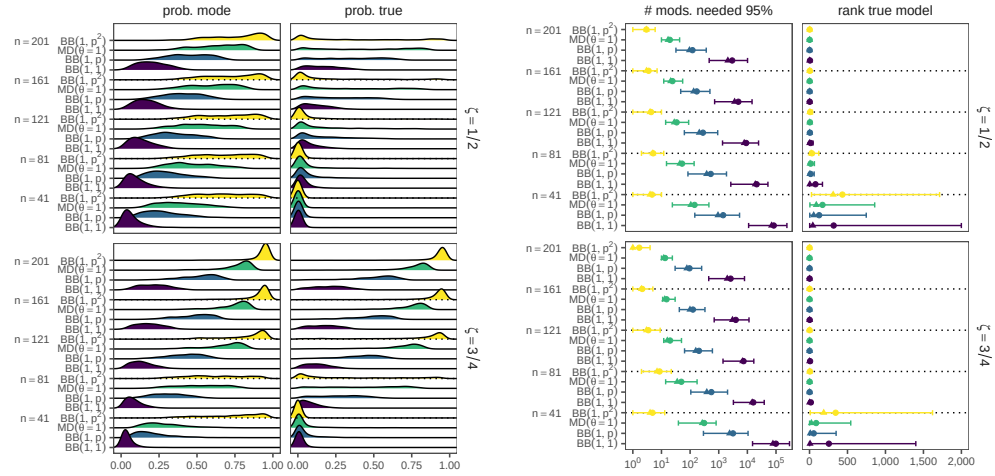


Figure 8: Posterior concentration metrics for simulation experiments over finite model spaces for growing n, fixed $p = 20$. The figure on the right displays the 95% confidence intervals, the mean (•) and the median (▲) for the number of models needed to accumulate 95% of the posterior distribution and for the rank of the true model.

The differences between the BB$(1, p^2)$, BB$(1, p)$, and MD$(\theta = 1)$ priors can be easily seen in both the probability of the modal model and the number of models needed for 95% of the probability mass of models visited by the search. The MD$(\theta = 1)$ places probability on the mode that is (on average) 1.8 times that of the BB$(1, p)$ and 0.8 that of the BB$(1, p^2)$. At the same time, the BB$(1, p)$ requires an average of 33 times the number of models of the MD$(\theta = 1)$ for posterior concentration and the MD$(\theta = 1)$ needs 60 times the number of models of the BB$(1, p^2)$. While the BB$(1, p^2)$ provides far too much certainty in a small number of models, the BB$(1, p)$ produces a posterior where uncertainty is spread across far too large of a set of models. The matryoshka doll once again strikes a compromise for between these two priors and produces reasonable posterior concentration.

7 Discussion

In this paper we propose a new construction for model space priors with a specific focus on regression models. The key motivation behind the design of the prior is universal multiplicity correction expressed through fixing the odds of a model versus the models that nest it. This provides the prior with both a meaningful interpretation and a construction from first principles that produces a well-balanced penalization profile across model (and model space) complexity. The matryoshka doll prior relies on a reformulation of the way in which the model space is viewed. Instead of considering indicator variables for covariate inclusion (as is typically done in the Bayesian paradigm), the

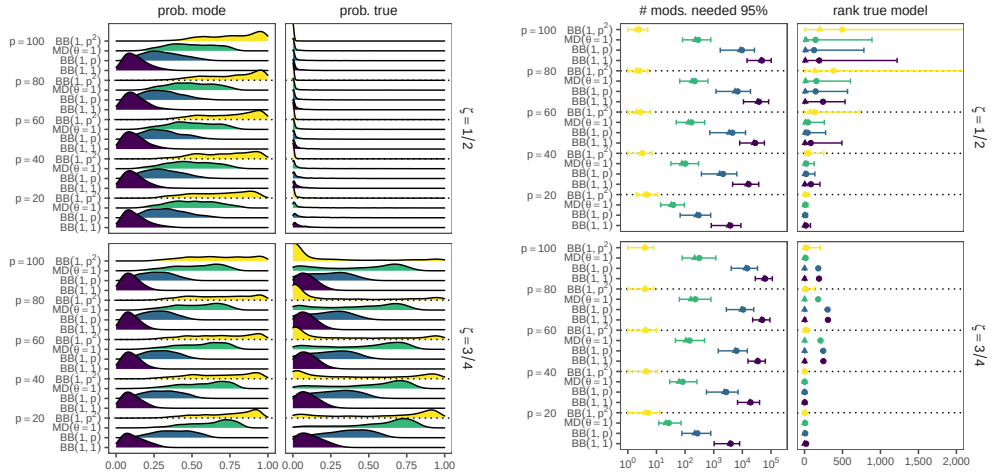


Figure 9: Posterior concentration metrics for simulation experiments over finite model spaces for growing p, fixed $n = 101$. The figure on the right displays the 95% confidence intervals, the mean (•) and the median (▲) for the number of models needed to accumulate 95% of the posterior distribution and for the rank of the true model.

model space is viewed as a series of local null and (sets of) alternative hypotheses. This induces the isomorphism theorem and the limiting Poisson prior on model complexity presented in Section 3. The generalization of the matryoshka doll class of priors, which arises from the most general statement of the isomorphism condition, shows the ubiquity of the Poisson distribution on model complexity when one focuses on multiplicity correction rather than on indicator variables for covariate inclusion.

We argue that explicit multiplicity correction (and induced complexity penalization) that strikes an attractive balance that eludes beta binomial priors is a better choice for constructing model space priors. The $\text{BB}(a, p^u)$ with $u > 1$ excessively controls false positives and thus has a tendency to omit true signals. Conversely, the $\text{BB}(a, b)$ or $\text{BB}(a, \lambda p)$ display weak control over the inclusion of false positive terms. The adaptation of the penalty to the number of covariates in the matryoshka doll provides both low penalization for adding the first few strong covariates to a model and high control of false positives when adding covariates that explain a small additional fraction of the variation in the response.

A final property of reformulating the prior construction in terms of local null and sets of alternative hypotheses is that the explicit multiplicity control readily generalizes to other model spaces that are described by posets. As a future direction, we are investigating the expansion of the matryoshka doll construction to more structured posets. In particular, we are interested in essentially non-parametric Bayesian models such as polynomial surface regression or multi-resolution analysis. The model spaces for these models can be forced to adhere to either weak or strong heredity rules (Taylor-Rodriguez

et al., 2016). For a fixed maximal degree (or subspace depth), the model space is a subset of $\mathcal{M}_p$, but the challenge lies in accommodating heredity requirements and the notion of a complexity class. As the size of the model space grows, difficulties arise in the asymptotic and self-similarity arguments analogous to those made in Section 4 for computing the matryoshka doll prior on these complex model spaces.

Funding

Womack was partially supported by NIH award 5R01DC018813-04.

Taylor-Rodríguez was partially supported by NSF RTG DMS award 2136228, NIH award 5R01DC018813-04, and by Portland State University's FDG and Sabbatical programs.

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Appendix A: Summary tables Section 6 simulations

Fixed	Varying	Prior	Prob Mode	Prob M_T	Rank M_T	mods. needed
$\zeta = 1/2$, $p = 20$	$n = 41$	BB(1,p)	0.26	0	55	943.5
		MD($\theta = 1$)	0.38	0	89	106
		BB(1,p^2)	0.66	0	311.5	4
	$n = 81$	BB(1,p)	0.29	0.03	5	379.5
		MD($\theta = 1$)	0.45	0.01	7	44
		BB(1,p^2)	0.68	0	21	5
	$n = 121$	BB(1,p)	0.34	0.11	2	209.5
		MD($\theta = 1$)	0.52	0.08	3	28
		BB(1,p^2)	0.71	0.01	5	4
	$n = 161$	BB(1,p)	0.40	0.25	1	133
		MD($\theta = 1$)	0.61	0.26	2	21
		BB(1,p^2)	0.74	0.07	2	3
	$n = 201$	BB(1,p)	0.45	0.35	1	95
		MD($\theta = 1$)	0.66	0.44	1	17
		BB(1,p^2)	0.80	0.22	2	3
$\zeta = 1/2$, $n = 101$	$p = 20$	BB(1,p)	0.3	0.1	3	253
		MD($\theta = 1$)	0.5	0	4	33
		BB(1,p^2)	0.7	0	7	4
	$p = 40$	BB(1,p)	0.3	0	5	1623.5
		MD($\theta = 1$)	0.5	0	6	78
		BB(1,p^2)	0.7	0	29	3
	$p = 60$	BB(1,p)	0.3	0	8	3481
		MD($\theta = 1$)	0.5	0	12	131
		BB(1,p^2)	0.8	0	74	2
	$p = 80$	BB(1,p)	0.3	0	10.5	5560.5
		MD($\theta = 1$)	0.5	0	15	174.5
		BB(1,p^2)	0.8	0	140	2
	$p = 100$	BB(1,p)	0.3	0	13	8453
		MD($\theta = 1$)	0.5	0	19	227
		BB(1,p^2)	0.8	0	201	2

Table 3: Medians of posterior concentration measures over 1000 simulated datasets from simulations in Section 6, having signal decay rate $\zeta = 1/2$ as either n or p grow. $\eta = 1/(\mathrm{e} - 1)$ so that $\theta = 1$ in matryoshka doll.

Fixed	Varying	Prior	Prob Mode	Prob M_T	Rank M_T	mods. needed
$\zeta = 3/4$, $p = 20$	$n = 41$	BB(1,p)	0.18	0.01	10	2396
		MD($\theta = 1$)	0.26	0	24.5	266
		BB(1,p^2)	0.77	0	187	4
	$n = 81$	BB(1,p)	0.31	0.28	1	383.5
		MD($\theta = 1$)	0.53	0.45	1	35.5
		BB(1,p^2)	0.61	0.2	2	7
	$n = 121$	BB(1,p)	0.44	0.43	1	158
		MD($\theta = 1$)	0.72	0.72	1	16
		BB(1,p^2)	0.88	0.87	1	3
	$n = 161$	BB(1,p)	0.5	0.5	1	99
		MD($\theta = 1$)	0.77	0.77	1	13
		BB(1,p^2)	0.94	0.94	1	2
	$n = 201$	BB(1,p)	0.54	0.54	1	77
		MD($\theta = 1$)	0.8	0.8	1	12
		BB(1,p^2)	0.95	0.95	1	1
$\zeta = 3/4$, $n = 101$	$p = 20$	BB(1,p)	0.4	0.4	1	218.5
		MD($\theta = 1$)	0.7	0.7	1	21
		BB(1,p^2)	0.8	0.7	1	4
	$p = 40$	BB(1,p)	0.3	0.3	1	2354.5
		MD($\theta = 1$)	0.6	0.6	1	61
		BB(1,p^2)	0.7	0.3	2	4
	$p = 60$	BB(1,p)	0.3	0.3	1	5596
		MD($\theta = 1$)	0.6	0.5	1	107
		BB(1,p^2)	0.7	0.1	2	4
	$p = 80$	BB(1,p)	0.3	0.2	1	9424
		MD($\theta = 1$)	0.5	0.5	1	160
		BB(1,p^2)	0.7	0	3	4
	$p = 100$	BB(1,p)	0.3	0.2	1	13034.5
		MD($\theta = 1$)	0.5	0.5	1	216.5
		BB(1,p^2)	0.7	0	4	4

Table 4: Medians of posterior concentration measures over 1000 simulated datasets from simulations in Section 6, having signal decay rate $\zeta = 3/4$ as either n or p grow. $\eta = 1/(\mathrm{e} - 1)$ so that $\theta = 1$ in matryoshka doll.

Appendix B: Model space exploration algorithm

In practice, when considering variable selection search algorithms are often preferred to exhaustive enumeration of the model space due to the computationally prohibitive cost the latter approach. Here we put forth a fast and effective strategy to search the space of models by building a binary tree with each model as a possible node.

Algorithm 1 performs model enumeration using a bifurcation and recursion algorithm. This algorithm provides a computationally efficient way to perform model space enumeration. To do so, it creates a total ordering out of the model space by starting at the least element, the base model (nested in all other models in the space), $M_\emptyset$, and taking the greediest path to the greatest element, the full model, M_F. It then goes back to the last place in the path, say M, where there are children models available that have not yet been chosen and sets the next model to be the best child of M that has not yet been visited. This is repeated until the model space is exhausted.

To set some notation, let $A \subseteq \{1, \ldots, p\}$, and define $Q(A)$ to be the unnormalized posterior probability of M_A, which is given by $Q(A) = f(\mathbf{y}|\mathbf{X}, M_A)\,\pi(M_A)$. This can be computed for any A whenever needed. Let Info(A) represent all of the information about model M_A that can be computed without enumerating the model space and that we want to write to file or memory. Furthermore, let U denote a set of terms that are not in M_A, which is initialized at $U = A^c$ whenever A is updated in the algorithm.

Algorithm 1: Greedy Binary Tree via Bifurcation and Recursion

```
Define: B&R (A ⊊ {1,...,p}, U ⊆ A^c)
    Set: u* = argmax_{u∈U} Q(A ∪ {u})
    Write: Info(A ∪ {u*})
    if |U| > 1 then
        Call: B&R (A ∪ {u*}, U \ {u*})
        Call: B&R (A, U \ {u*})
    end if
end B&R
Run:
    Write: Info(∅)
    Call: B&R (∅, {1,...,p})
end Run
```

In Algorithm 2, we perform the same bifurcation and recursion algorithm with the addition of a pruning step, limiting wasteful exploration of regions of the model space where little probability mass can be found. To implement this last step, we assume $\epsilon \in (0, 1)$, and the algorithm tests if the probability for a proposed model relative to its path is greater than ϵ to continue with the recursion, if it is not then the branch is pruned at that last model.

In addition to the definitions provided above, in Algorithm 2 the pruning step depends on the quantity q, which measures the probability that has been accumulated

from $M_\emptyset$ along the branch leading up to the model being considered. Note that q is initialized at $Q(\emptyset)$. As before, the algorithm proposes at each step the most advantageous available step from a model M_i, say $M_i^\star$. If there is enough evidence for $M_i^\star$, then it is added to the path as M_{i+1} and we continue the construction. Starting from $M_\emptyset$ and going along the branch, once the evidence in favor of some $M_i^\star$ is too small relative to the branch, the move is rejected and the descendants of M_i are pruned from the model space and removed from consideration. Once such pruning occurs, we move back down the branch to the most recently visited model with a children set that has not been completed visited and pruned. We then propose a best move from that model and evaluate its merit relative to its path from $M_\emptyset$.

Algorithm 2: Pruned Greedy Binary Tree via Bifurcation and Recursion

Define: PB&R $(A \subsetneq \{1,\ldots,p\},\ U \subseteq A^c, q > 0)$
  Set: $u^* = \text{argmax}_{u \in U}\, Q\left(A \cup \{u\}\right)$
  Write: Info$(A \cup \{u^*\})$
  if $|U| > 1$ **then**
    if $\frac{Q(A \cup \{u^*\})}{(q + Q(A \cup \{u^*\}))} > \epsilon$ **then**
      Call: PB&R $(A \cup \{u^*\}, U \setminus \{u^*\}, q + Q\left(A \cup \{u^*\}\right))$
    end if
    Call: PB&R $(A, U \setminus \{u^*\}, q)$
  end if
end PB&R
Run:
  Write: Info$(\emptyset)$
  Call: PB&R $(\emptyset, \{1,\ldots,p\}, Q(\emptyset))$
end Run

Supplementary Material

Appendix C: Algorithm performance evaluation

To validate the proposed algorithm, we make use of the same set of simulations presented in Section 6 of the main body of paper and the compare results from enumeration (generated using Algorithm 1) to those obtained using Algorithm 2, and derive three metrics. First, the total variation norm (tv norm) between the probabilities of models found by the search algorithm and the probabilities for those same models obtained from enumeration. The second metric considered is the total probability mass discovered by the search (prob. found search), which was obtained by adding together the *true* model posterior probabilities (i.e., those from enumeration) of models found by the search algorithm. Finally, we extracted the maximum posterior probability among models not discovered by the algorithm (max prob. missed).

The purpose of this exercise is two-fold. First, we want to show the behavior of

the algorithm. Second, we want to show the behavior of the priors. In particular, the $\text{BB}(a, \lambda p)$ prior does not penalize complexity enough. Its posterior does not adequately concentrate on a set of "good" models due to poor control of prior odds to nesting or children models as true model dimension grows (see Section 5 of the paper).

The two panels in Figure 10 summarize the results obtained for simulations with growing n and growing p, respectively. We find that the model space prior considered influences the effectiveness of the algorithm, which is to be expected given that the posterior concentration strongly depends on the model space prior. The general patterns we find are:

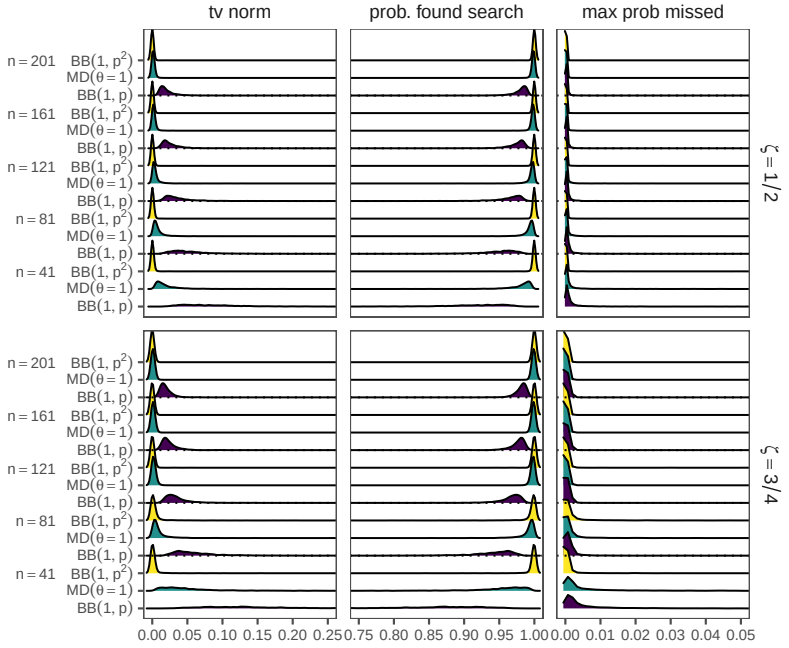
1. In terms of the tv norm, the $\text{BB}(1, p^2)$ is always closest to zero, meaning that the renormalized probabilities obtained with the models found by the algorithm closely match their true values, and that the bulk of the probability mass is found. The $\text{MD}(\theta = 1)$ follows closely, also being close to zero over most settings while being shifted slightly to the right with respect to the $\text{BB}(1, p^2)$. Finally, under the $\text{BB}(1, p)$ the tv norm lie slightly further away from zero. This pattern based on the prior makes sense since it is impacted by how each prior distributes the probability mass over models, with the $\text{BB}(1, p^2)$ concentrating the mass over only a few models while the $\text{BB}(1, p)$ comparatively dilutes it over the space. Differences in the behavior of the tv norm across distributions are most evident in the scenarios with growing p.

2. For prob. found search we find that the $\text{BB}(1, p^2)$ consistently finds most of the probability mass (i.e., it is always close to one), followed slightly lower values from the $\text{MD}(\theta = 1)$, which is then followed by lower values (or drastically lower values depending of the scenario) by the $\text{BB}(1, p)$. This result follows the same intuition as with the tv norm, that is, this result is a consequence of the different posterior concentration regimes provided by the different priors.

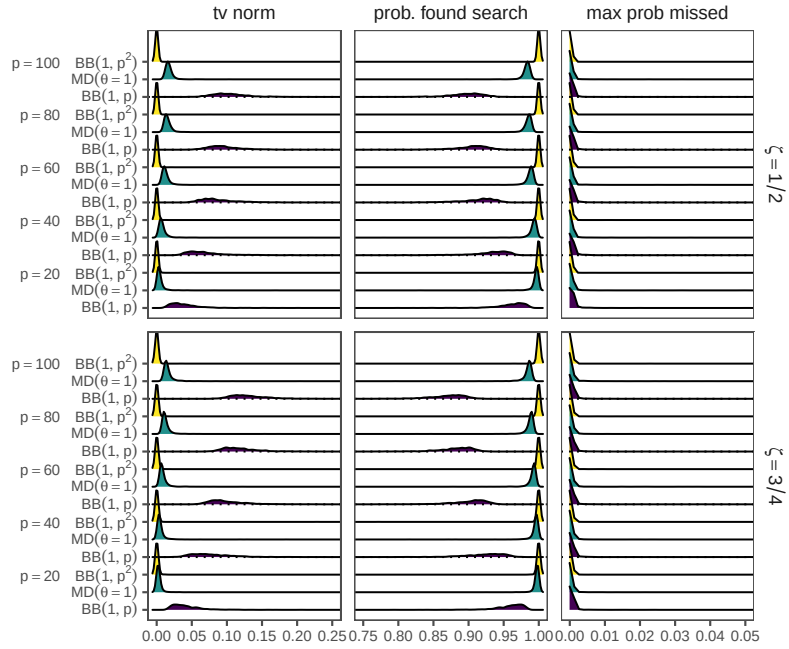
3. Over all scenarios considered there is barely any difference across priors in the max prob. missed. This metric is invariably close to zero, meaning that the algorithm is able to find the vast majority or all high probability models regardless of the prior.

Appendix D: Prior behavior on growing model spaces

In this section we explore the behavior of the matryoshka doll prior through simulations and provide empirical support for the claims made in previous sections. We compare the matryoshka doll prior ($\text{MD}(\theta = 1)$, to the $\text{BB}(1, b)$ prior with $b \in \{1, p, p^2\}$. We evaluate the posterior behavior in terms of signal reconstruction and posterior concentration. Both the number of possible predictors as well as the number of predictors in the true signal increase with sample size. The predictors are generated as independent draws from a standard normal distribution and the response is normal with the appropriate mean structure and variance $\sigma^2 = 1$. We evaluate: (i) posterior behavior, both with finite samples and asymptotically as the number of predictors grows with n; and (ii) the



(a) Growing n, fixed $p = 20$



(b) Growing p, fixed $n = 101$

Figure 10: Algorithm efficacy metrics for simulation experiments over finite model spaces as either the same size n or the total number of predictors in the full model p grow.

influence of having the signal in the predictors decay rapidly (i.e., a large proportion of signal in the true model concentrates in a few of the predictors) or slowly (i.e., signal in true model is more evenly spread out across all predictors).

The sample sizes considered are $n = 200, 2000$. The size of the full model M_F is $p = 2n$ and the size of the true model M_T is $|M_T| \approx \sqrt{n}$. We assume that $M_T \subset M_F$. To control how the signal is distributed across all predictors in M_T, we assume its jth largest regression coefficient β_j satisfies $\beta_j^2 \propto \zeta^j$ with $\zeta \in \{1/2, 9/10\}$. We assume that the asymptotic mean structure is generated from the full geometric series with signal to noise ratio $\sum \beta_j^2/\sigma^2 = 2$. The mean structure for M_T for finite n is generated by the series truncated to the appropriate number of terms. With each combination of n and ζ we simulated 100 datasets.

Setting the decay rate $\zeta = 1/2$ makes the magnitude of the coefficients decrease rapidly and concentrate 99% of the true signal in seven regression coefficients. In contrast, setting $\zeta = 9/10$ promotes a slow decay, spreading out the signal more evenly across all predictors and requiring forty four predictors to account for 99% of the total signal.

The model space exploration algorithm described in Section 6 was driven by model posterior probabilities obtained with Zellner-Siow priors for the model parameters and the BB$(1, 1)$ prior for models. The branch pruning cutoff was set to $\varepsilon = 0.01$. These choices enabled the algorithm to visit a large portion of the model space because the BB$(1, 1)$ prior provides the weakest complexity penalization among all the priors considered. The posterior probabilities resulting for the other model space priors were calculated using the Bayes' factors obtained from the search and renormalization.

We evaluate the model space priors in terms of *false positive inclusion*, *true positive inclusion*, *model complexity*, and *posterior concentration*. The specific metrics used are described in the corresponding section of the analysis of the simulation study.

D.1 False and true positive rates

The false and true positive rate metrics are both discrete and continuous.

1. False positive rates:

$$\begin{aligned} \mathsf{FPR}_{\mathsf{discr}} &= \sum_{M \in \mathcal{M}} \mathsf{fpr}_{\mathsf{discr}}(M) \cdot p(M|\mathbf{y}), \\ \mathsf{FPR}_{\mathsf{cont}} &= \sum_{M \in \mathcal{M}} \mathsf{fpr}_{\mathsf{cont}}(M) \cdot p(M|\mathbf{y}), \text{ with} \end{aligned}$$

$$\mathsf{fpr}_{\mathsf{discr}}(M) = \frac{|M \setminus M_T|}{|M_T|} \quad \text{and} \quad \mathsf{fpr}_{\mathsf{cont}}(M) = 1 - \frac{R^2_{M \cap M_T}}{R^2_M},$$

where M_T represents the true model, $M \setminus M_T$ corresponds to the set of terms in M but not in M_T, $M \cap M_T$ denotes the set of terms in both M and M_T, and finally, R^2_A represents the unadjusted R-squared for M_A.

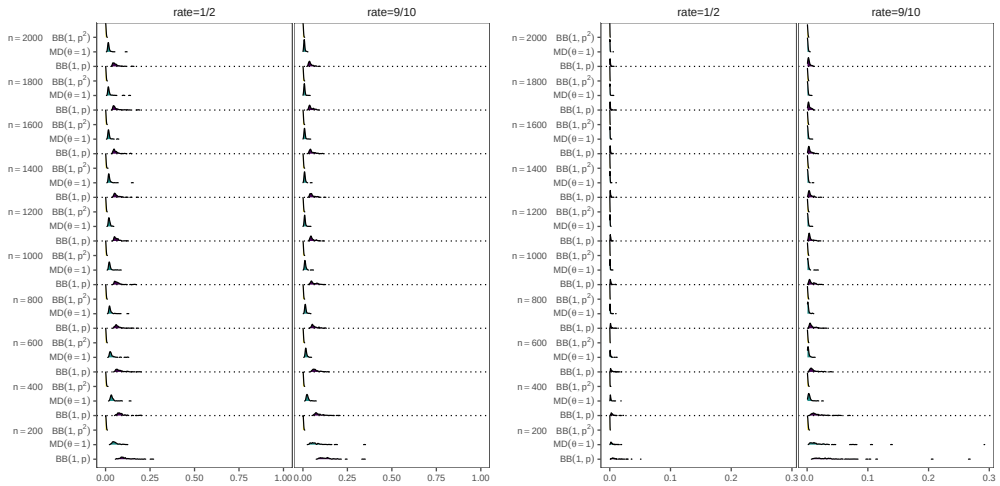
2. True positive rates:

$$\begin{aligned} \mathsf{TPR}_{\mathsf{discr}} &= \sum_{M \in \mathcal{M}} \mathsf{tpr}_{\mathsf{discr}}(M) \cdot p(M|\mathbf{y}), \\ \mathsf{TPR}_{\mathsf{cont}} &= \sum_{M \in \mathcal{M}} \mathsf{tpr}_{\mathsf{cont}}(M) \cdot p(M|\mathbf{y}), \text{ with} \end{aligned}$$

$$\mathsf{tpr}_{\mathsf{discr}}(M) = \frac{|M \cap M_T|}{|M_T|} \quad \text{and} \quad \mathsf{tpr}_{\mathsf{cont}}(M) = \frac{R^2_{M \cap M_T}}{R^2_{M_T}}.$$

The discrete and a continuous versions for the FPR and TPR convey different information about signal reconstruction. The discrete versions inform about the number of (incorrect or correct) predictors selected, while the continuous version provides insights about the amount of signal explained (by the false or true predictors). A small continuous FPR need not correspond to a small discrete FPR. Similarly, a large continuous TPR need not correspond to a large discrete TPR.

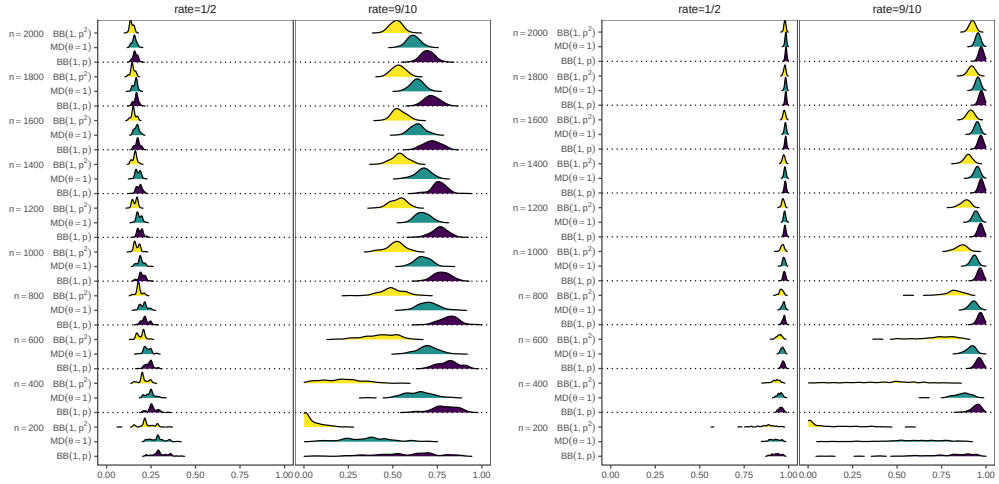
Overall, the results observed in terms of FPR and TPR follow the expected patterns (see Figures 11 and 12), implying a tradeoff between the FPR and TPR. The resulting metrics are neatly ordered, with those for the $\text{BB}(1, p^2)$ sitting at one extreme, followed by the matryoshka doll, then the $\text{BB}(1, p)$ and the $\text{BB}(1, 1)$ at the opposite end of the spectrum.



(a) Discrete FDR (b) Continuous FDR

Figure 11: Expected false discovery rates. Sample sizes $n = 200, 400, \ldots, 2000$ and decay rates of true coefficient signal strength $\zeta = 1/2$ and $\zeta = 9/10$.

In terms of the FPR (Figure 11), both for the continuous and discrete versions, the $\text{BB}(1, p^2)$ is consistently near zero regardless of the sample size or the signal decay rate, which is expected as this prior yields the strongest penalization. The matryoshka doll prior takes only slightly higher values than the $\text{BB}(1, p^2)$ in both $\mathsf{FPR}_{\mathsf{discr}}$ and $\mathsf{FPR}_{\mathsf{cont}}$. The $\mathsf{FPR}_{\mathsf{discr}}$ values for the matryoshka doll prior are significantly smaller than those for the $\text{BB}(1, p)$ and $\text{BB}(1, 1)$ priors. Finally, regardless of the prior, the values observed for $\mathsf{FPR}_{\mathsf{cont}}$ concentrate about 0, indicating that the signal added by the false predictors is negligible.



(a) Discrete TPR

(b) Continuous TPR

Figure 12: Posterior expected true positive rates. Sample sizes $n = 200, 400, \ldots, 2000$ and decay rates of true coefficient signal strength $\zeta = 1/2$ and $\zeta = 9/10$.

Conversely, the results obtained for the true positive rates (Figure 12) shed light on a fascinating interplay between the model space prior, the sample size and the signal decay rate, with the discrete and continuous versions telling different parts of the story. With a $\zeta = 1/2$ decay rate, the distribution of the TRPs (both discrete and continuous) across all priors are similar. The discrete version actually decreases as a function of n because most of the true signal concentrates on a few predictors, whereas the continuous TPR values increase towards one as n increases. This implies that with a rapid signal decay across true predictors, regardless of the prior considered, the true predictors left out correspond to those with low signal. There are minor differences between the model space priors when the sample size is small and the differences essentially disappear when the sample size is large.

When $\zeta = 9/10$ and the signal is more evenly spread out across predictors, the choice of model space prior impacts the inference, with the strength of its influence mediated by the sample size. With a sample size of $n = 200$, the $\text{BB}(1, p^2)$ prior severely penalizes the addition of covariates into the model and the $\text{BB}(1, 1)$ and $\text{BB}(1, p)$ allow

in the most true predictors. As expected, the matryoshka doll produces a TPR that sits between these two extremes. With $n = 2000$, the same ordering remains but the differences across all priors become considerably less pronounced, especially in terms of the signal (see panel (b) in Figure 12). In terms of the number of true predictors, the simulation density of discrete TPR for the matryoshka doll overlaps with those of both the $\text{BB}(1, p^2)$ and the $\text{BB}(1, p)$; however, the latter distributions display little overlap among themselves (Figure 12 panel (a)). In terms of the continuous TPR, the same ordering appears. Notably, the $\text{BB}(1, p^2)$ prior's extreme control of false positives is driving down the continuous TPR and meaningful variation in the data from covariates in the true model is being missed.

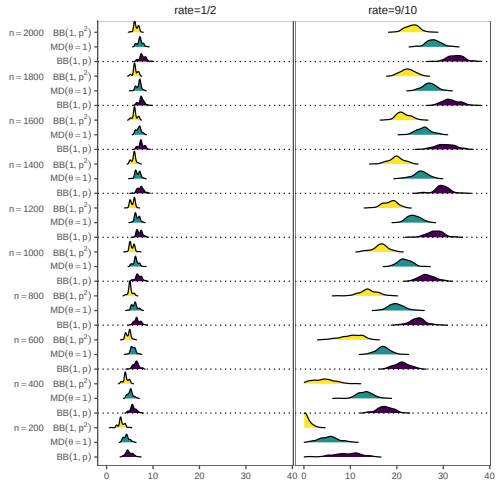
D.2 Posterior concentration

Finally, to evaluate the posterior concentration we assess three metrics: posterior expected model complexity, posterior probability of the modal model, and the number of models needed to accumulate 95% of the total probability mass of the models found by the search algorithm. These are shown in Figure 13 and medians are presented in Table 5. Together with true and false positive rates, these metrics help to provide a more complete understanding of the behaviors of the priors. The two cases for ζ reflect different expectations in posterior behavior.

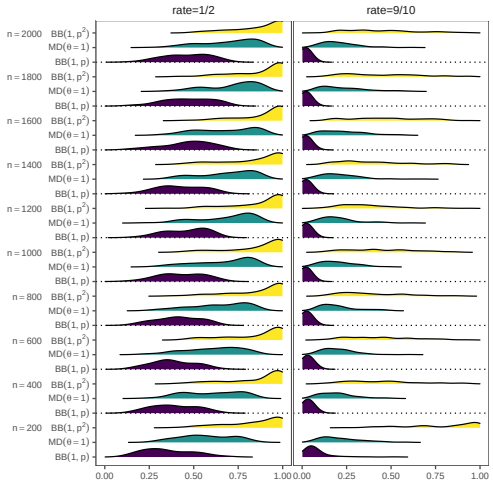
		$n = 200$			$n = 2000$		
Rate	Prior	Comp	Conc	Max	Comp	Conc	Max
$\zeta = \frac{1}{2}$	$\text{BB}(1, 1)$	5.4	3364	0.179	8.09	9466	0.292
	$\text{BB}(1, p)$	4.7	1698	0.332	7.64	5562	0.47
	$\text{MD}(\eta)$	4.26	422	0.565	7.15	651	0.746
	$\text{BB}(1, p^2)$	3.02	2	0.887	6.16	2	0.93
$\zeta = \frac{9}{10}$	$\text{BB}(1, 1)$	12.13	30830	0.016	34.4	390398	0.006
	$\text{BB}(1, p)$	9.54	29168	0.053	32.66	389586	0.017
	$\text{MD}(\eta)$	5.49	3529	0.188	27.91	14738	0.183
	$\text{BB}(1, p^2)$	0.16	2	0.871	23.28	10	0.414

Table 5: Medians of measures over 100 simulated datasets. $\eta = 1/(\text{e} - 1)$ so that $\theta = 1$ in matryoshka doll. Comp is posterior expected model complexity. Conc is the number of models needed to comprise 95% of the model space posterior (rounded to whole numbers). Max is the posterior probability of the modal model.

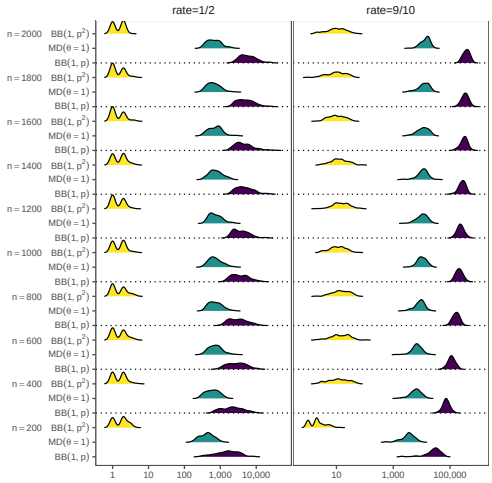
First, we discuss the case when $\zeta = 1/2$ and the true model is well approximated by seven predictors. The $\text{BB}(1, p^2)$ prior provides an unwarranted amount of certainty (or lack of uncertainty) in regards to what the "true" model is and what constitutes the set of good models. The excessively high modal posterior model probability and the posterior concentration on a few models reflects this prior's unnecessarily severe penalization of false positives. On the other end of the spectrum is the $\text{BB}(1, 1)$ and $\text{BB}(1, p)$ model, which have a higher false positive rate and thus concentrate less in



(a) Model Complexity



(b) Maximum probability



(c) Models needed for 95% probability mass

Figure 13: Posterior average model complexity, modal posterior model probability, and number of models needed to accumulate 95% of the total probability mass. Sample sizes $n = 200$ and $n = 2000$ and decay rates of true coefficient signal strength $\zeta = 1/2$ and $\zeta = 9/10$.

the posterior. This causes a higher average true positive rate, but also implies that thousands of models are needed to acummulate 95% of the model space posterior mass. As expected, the matryoshka doll sits between these two extremes. It finds the most important predictors and concentrates on hundreds and not thousands of models. This posterior model space uncertainty is strongly contrasted with the certainty obtained by the $\text{BB}(1, p^2)$, which concentrates on one or two models.

Second, we discuss the case when $\zeta = 9/10$ and the model needs a large number of predictors to capture the mean structure. Especially when n is small, the $\text{BB}(1, p^2)$ provides an unjustified amount of concentration on a small number of models. In contrast, the $\text{BB}(1, 1)$ and $\text{BB}(1, p)$ spread the probability mass across too many models and assign low posterior probability to their modal models. As expected, the matryoshka doll provides a data driven compromise between these two extremes. For example, when $n = 2000$, its concentration on tens of thousands of models, as opposed to tens of models for $\text{BB}(1, p^2)$ and hundreds of thousands for $\text{BB}(1, 1)$ and $\text{BB}(1, p)$, is reasonable given the complexity of its modal model and the number of covariates left that explain meaningful variation in the response variable. The average posterior complexity for the matryoshka doll prior is about 28 and we need 44 covariates to explain 99% of the variation in the response. This leads to tens of thousands of models that are nested in the true model and that nest the modal model. It is completely reasonable to have the posterior express this amount of uncertainty in what the true model is when $\zeta = 9/10$.

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